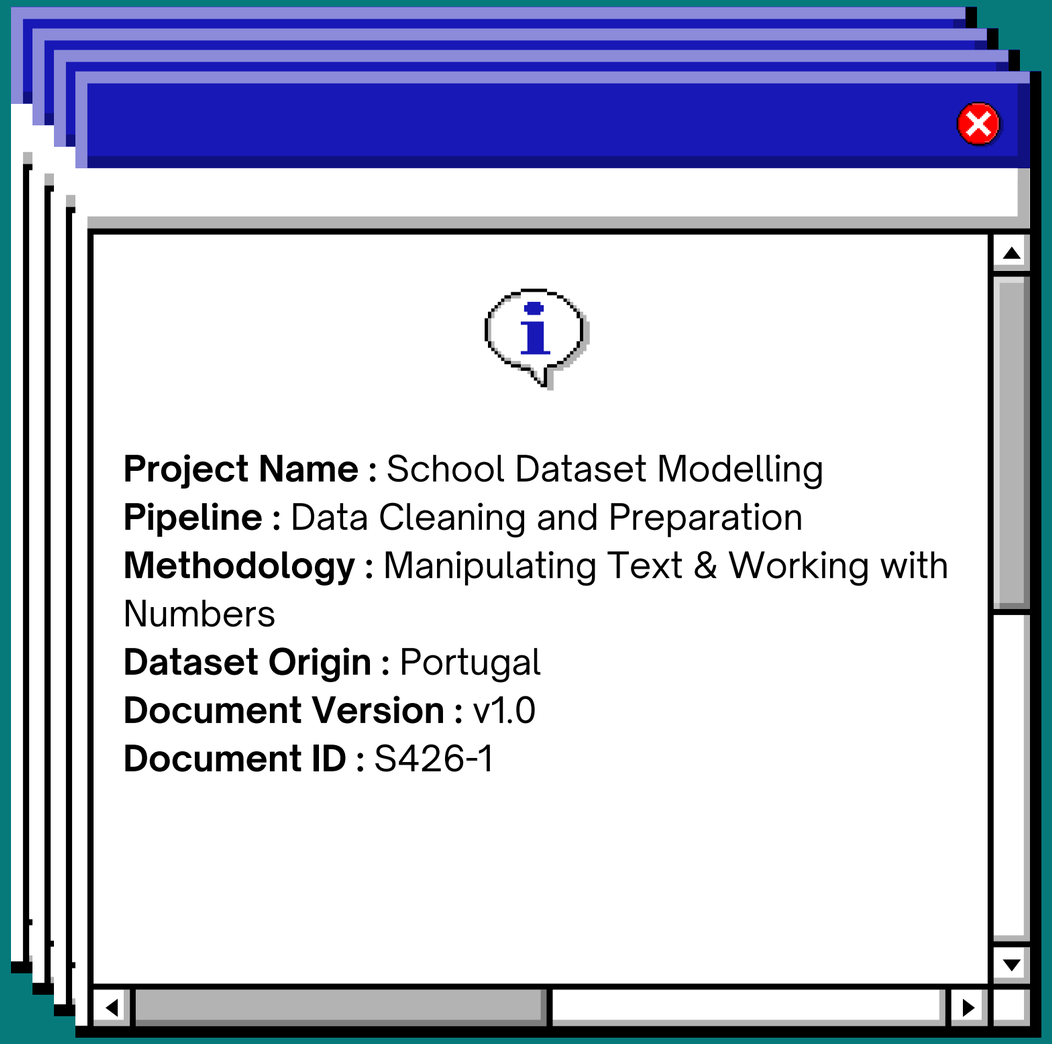
School Performance & Demographic Data Analysis

School Grades & Demographic Dataset



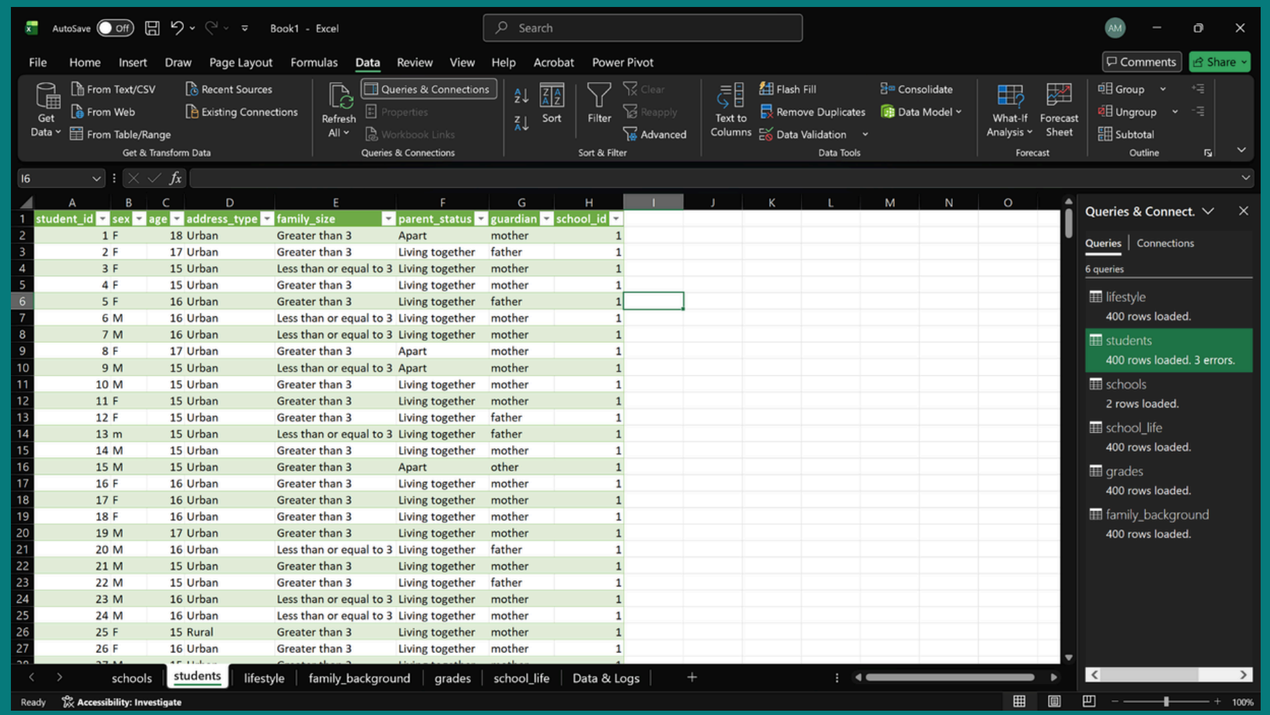
Project Name : School Dataset Modelling
Pipeline : Data Cleaning and Preparation
Methodology : Manipulating Text & Working with Numbers
Dataset Origin : Portugal
Document Version : v1.0
Document ID : S426-1

Data Cleaning & Preparation

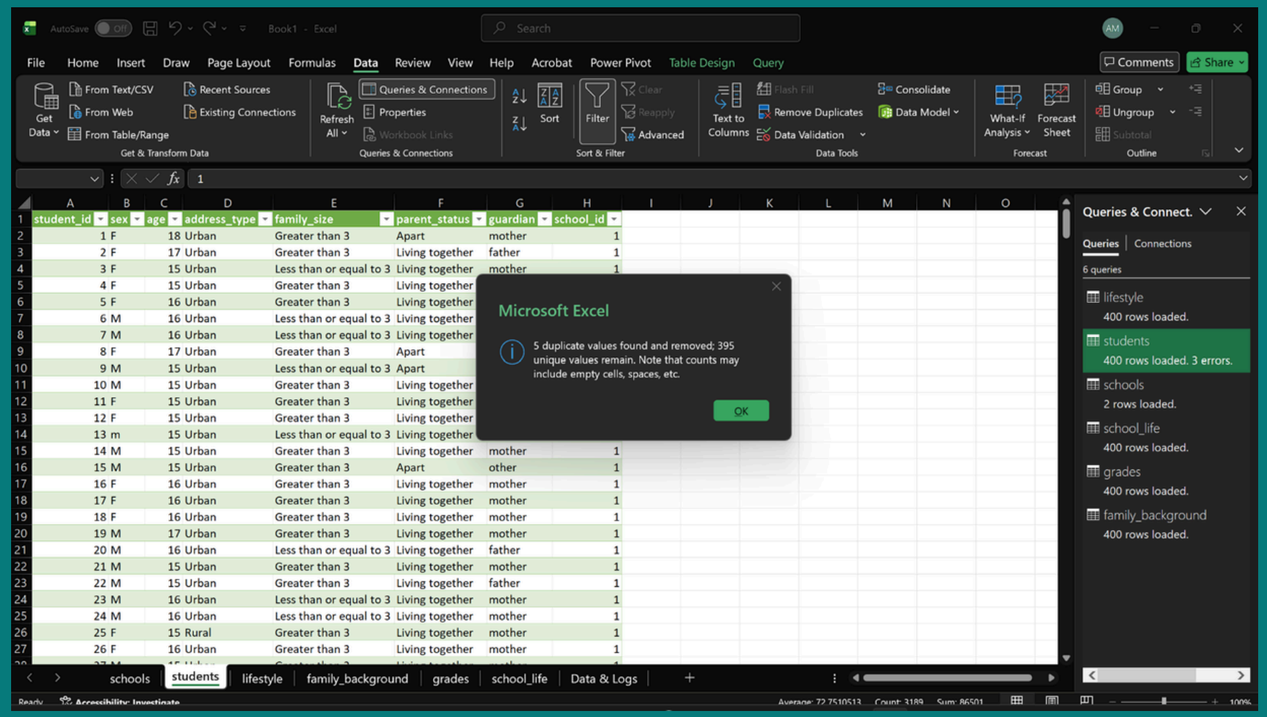
Corrected Order of Operations			
Order	**Step**	**Process**	**Columns Affected**
01	Remove Duplicates	1.Data Tab 2.Remove Duplicates	All Tables
02	TRIM Whitespace	1. Make Helper Column 2. [ALL]=Trim(Value) 3. Copy Paste New Values 4. Delete Helper Column	school_choice_reason guardian, mother_job, father_job
03	Standardize Capitalization	1. Make Helper Column 2.[sex]=IF(UPPER(LEFT(TRIM(B2),1))="F","F","M") 3. Copy Paste New Values 4.Delete Helper Column	sex
04	Standardize Yes/No	1.Make Helper Column 2.[ALL]=IF(LOWER(LEFT(TRIM(E2),1))="y","yes","no") 3.Copy Paste New Values 4.Delete Helper Column	school_support, family support, extra_paid_classes, activities, internat_access
05	Fix Mixed Data Types	1.Make Helper Column 2.=IF(OR(G2="unknown",G2=""),17,VALUE(G2)) 3.Copy Paste New Values 4.Delete Helper Column	age
06	Fix Out of Range Values	1.Make Helper Column 2.[grade]=IF(OR(D2<0,D2>20),AVERAGE(B2,C2),D2) 3.[absences]=IF(K2<0,0,K2) 4.Copy Paste New Values 5.Delete Helper Column	final_grade, absences
07	Impute Missing Values	1.Make Helper Column 2.[health]=MEDIAN(lifestyle!J2:J396) 3.[health]=IF(ISBLANK(J2),4,J2) 4.[absences]=IF(ISBLANK(K2),0,K2) 5.[grade_1]=IF(ISBLANK(B2),AVERAGE(C2,D2),B2) 6.[grade_2]=IF(ISBLANK(C2),AVERAGE(B2,D2),C2) 7.[final_grade]=IF(ISBLANK(D2),AVERAGE(B2,C2),D2) 8.Delete Helper Column	age(already), health, absences, grade_1, grade_2, final_grade

Remove Duplicate Rows

Before (Raw)



After (Clean)



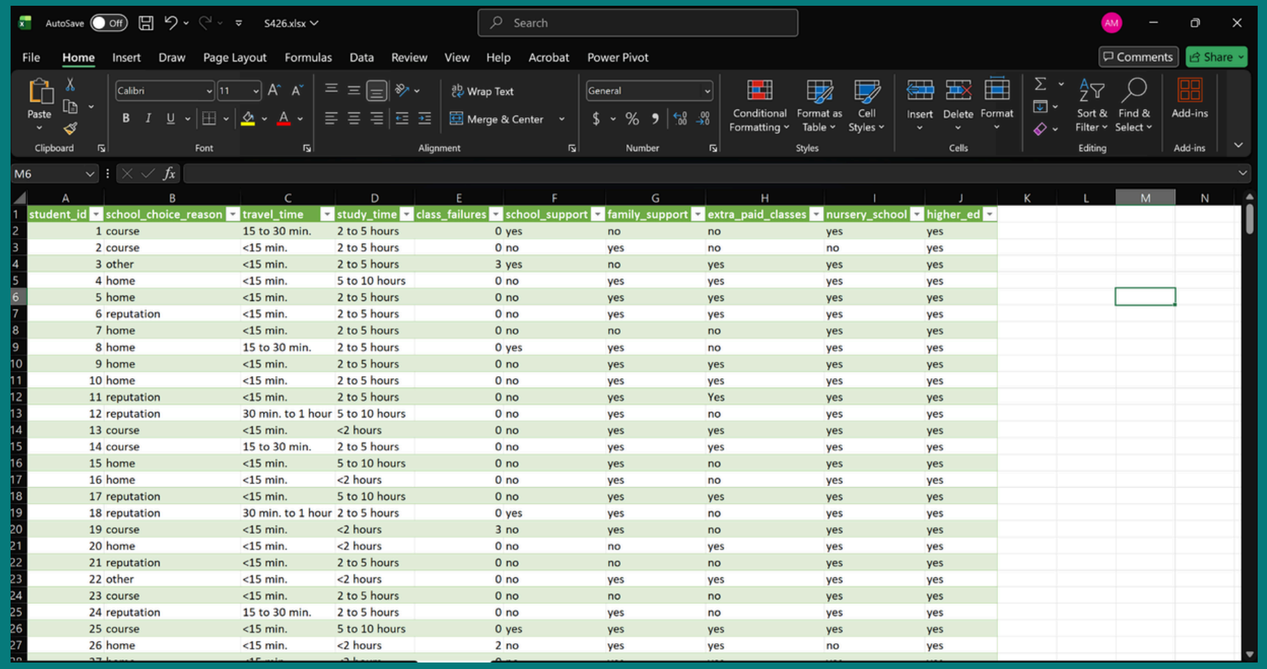
Process : 1.Data Tab
2.Remove Duplicates

Columns Affected :

- Students - 5 Duplicates Found & Removed
- All Table

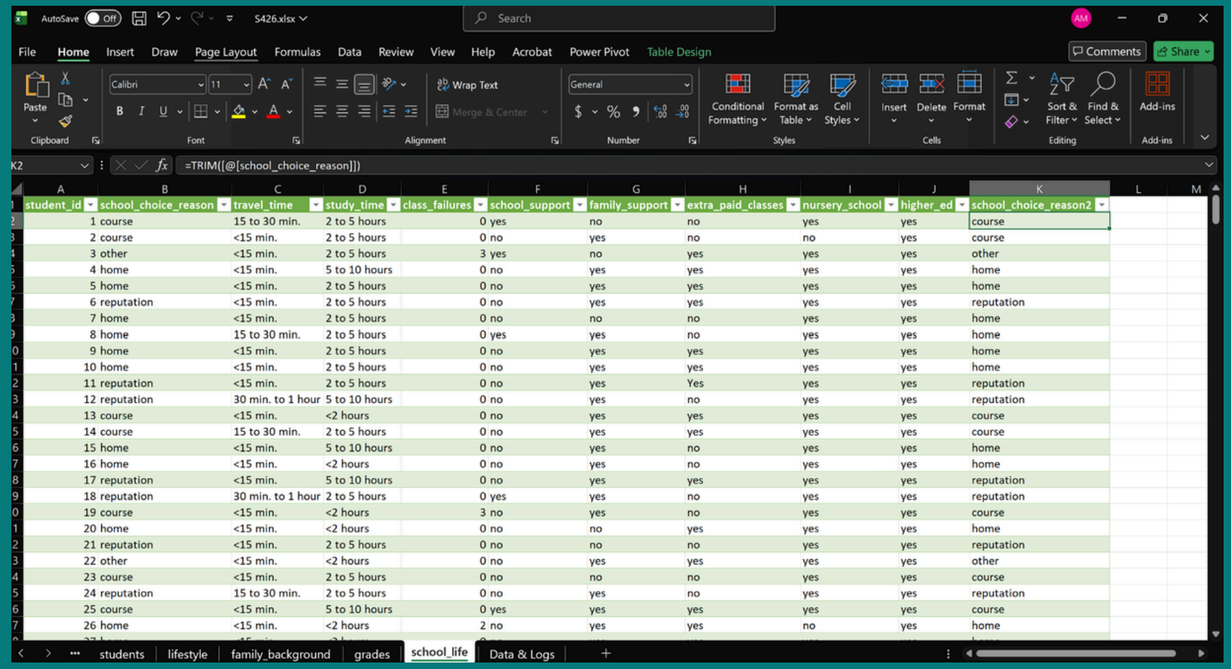
TRIM Whitespace

Before (Raw)



student_id	school_choice_reason	travel_time	study_time	class_failures	school_support	family_support	extra_paid_classes	nursery_school	higher_ed
1	course	15 to 30 min.	2 to 5 hours	0	yes	no	no	yes	yes
2	course	<15 min.	2 to 5 hours	0	no	yes	no	no	yes
3	other	<15 min.	2 to 5 hours	3	yes	no	yes	yes	yes
4	home	<15 min.	5 to 10 hours	0	no	yes	yes	yes	yes
5	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes
6	reputation	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes
7	home	<15 min.	2 to 5 hours	0	no	no	no	yes	yes
8	home	15 to 30 min.	2 to 5 hours	0	yes	yes	no	yes	yes
9	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes
10	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes
11	reputation	<15 min.	2 to 5 hours	0	no	yes	Yes	yes	yes
12	reputation	30 min. to 1 hour	5 to 10 hours	0	no	yes	no	yes	yes
13	course	<15 min.	<2 hours	0	no	yes	yes	yes	yes
14	course	15 to 30 min.	2 to 5 hours	0	no	yes	yes	yes	yes
15	home	<15 min.	5 to 10 hours	0	no	yes	no	yes	yes
16	home	<15 min.	<2 hours	0	no	yes	no	yes	yes
17	reputation	<15 min.	5 to 10 hours	0	no	yes	yes	yes	yes
18	reputation	30 min. to 1 hour	2 to 5 hours	0	yes	yes	no	yes	yes
19	course	<15 min.	<2 hours	3	no	yes	no	yes	yes
20	home	<15 min.	<2 hours	0	no	no	yes	yes	yes
21	reputation	<15 min.	2 to 5 hours	0	no	no	no	yes	yes
22	other	<15 min.	<2 hours	0	no	yes	yes	yes	yes
23	course	<15 min.	2 to 5 hours	0	no	no	no	yes	yes
24	reputation	15 to 30 min.	2 to 5 hours	0	no	yes	no	yes	yes
25	course	<15 min.	5 to 10 hours	0	yes	yes	yes	yes	yes
26	home	<15 min.	<2 hours	2	no	yes	yes	no	yes

After (Clean)



student_id	school_choice_reason	travel_time	study_time	class_failures	school_support	family_support	extra_paid_classes	nursery_school	higher_ed	school_choice_reason2
1	course	15 to 30 min.	2 to 5 hours	0	yes	no	no	yes	yes	course
2	course	<15 min.	2 to 5 hours	0	no	yes	no	no	yes	course
3	other	<15 min.	2 to 5 hours	3	yes	no	yes	yes	yes	other
4	home	<15 min.	5 to 10 hours	0	no	yes	yes	yes	yes	home
5	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes	home
6	reputation	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes	reputation
7	home	<15 min.	2 to 5 hours	0	no	no	no	yes	yes	home
8	home	15 to 30 min.	2 to 5 hours	0	yes	yes	no	yes	yes	home
9	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes	home
10	home	<15 min.	2 to 5 hours	0	no	yes	yes	yes	yes	home
11	reputation	<15 min.	2 to 5 hours	0	no	yes	Yes	yes	yes	reputation
12	reputation	30 min. to 1 hour	5 to 10 hours	0	no	yes	no	yes	yes	reputation
13	course	<15 min.	<2 hours	0	no	yes	yes	yes	yes	course
14	course	15 to 30 min.	2 to 5 hours	0	no	yes	yes	yes	yes	course
15	home	<15 min.	5 to 10 hours	0	no	yes	no	yes	yes	home
16	home	<15 min.	<2 hours	0	no	yes	no	yes	yes	home
17	reputation	<15 min.	5 to 10 hours	0	no	yes	yes	yes	yes	reputation
18	reputation	30 min. to 1 hour	2 to 5 hours	0	yes	yes	no	yes	yes	reputation
19	course	<15 min.	<2 hours	3	no	yes	no	yes	yes	course
20	home	<15 min.	<2 hours	0	no	no	yes	yes	yes	home
21	reputation	<15 min.	2 to 5 hours	0	no	no	no	yes	yes	reputation
22	other	<15 min.	<2 hours	0	no	yes	yes	yes	yes	other
23	course	<15 min.	2 to 5 hours	0	no	no	no	yes	yes	course
24	reputation	15 to 30 min.	2 to 5 hours	0	no	yes	no	yes	yes	reputation
25	course	<15 min.	5 to 10 hours	0	yes	yes	yes	yes	yes	course
26	home	<15 min.	<2 hours	2	no	yes	yes	no	yes	home

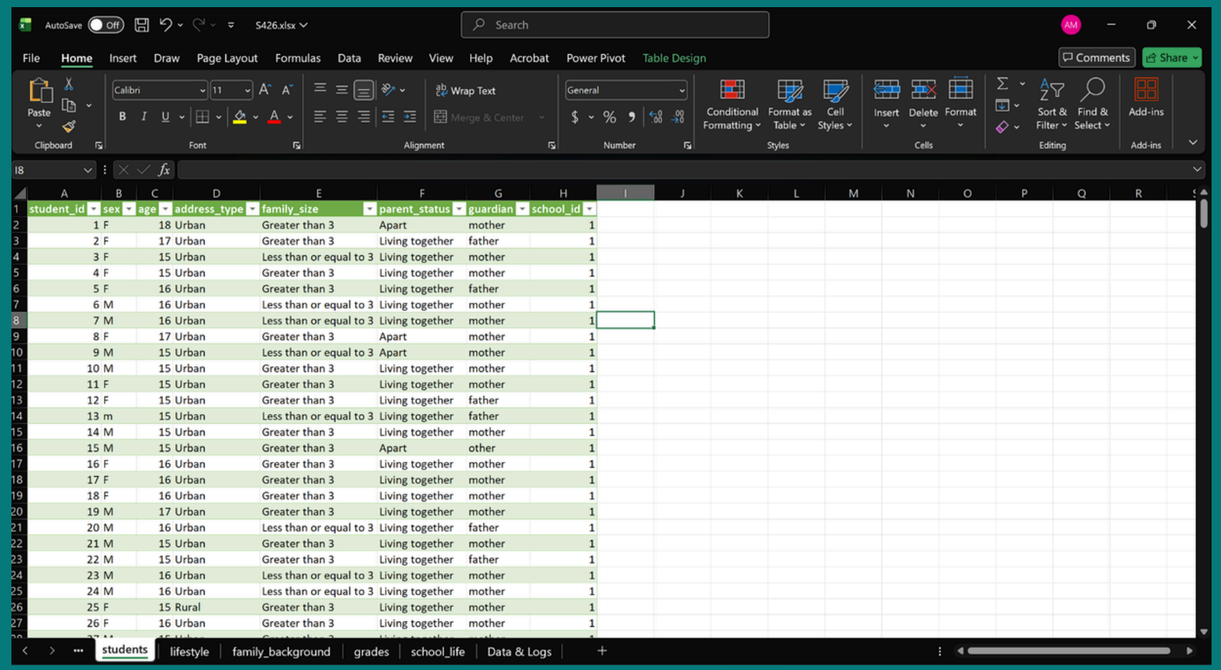
Process : 1. Make Helper Column
2. [ALL]=Trim(Value)
3. Copy Paste New Values
4. Delete Helper Column

Columns Affected :

- school_choice_reason(slide),
- guardian,
- mother_job,
- father_job

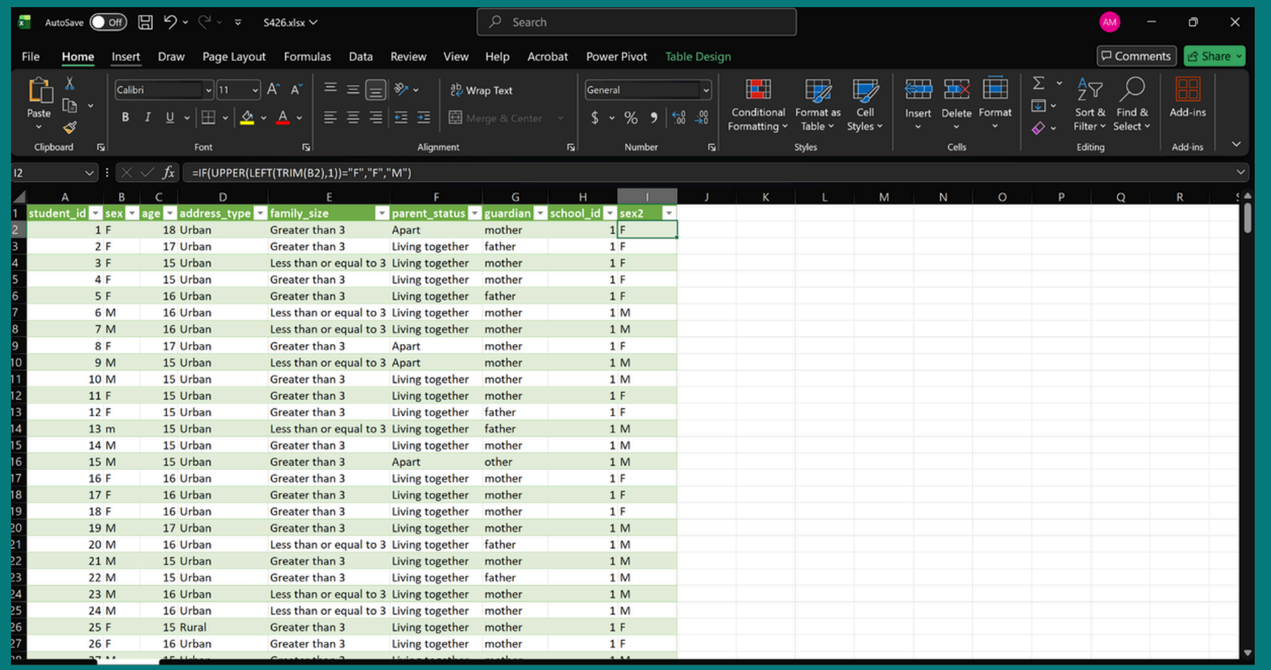
Standardize Capitalization

Before (Raw)



student_id	sex	age	address_type	family_size	parent_status	guardian	school_id
1	F	18	Urban	Greater than 3	Apart	mother	1
2	F	17	Urban	Greater than 3	Living together	father	1
3	F	15	Urban	Less than or equal to 3	Living together	mother	1
4	F	15	Urban	Greater than 3	Living together	mother	1
5	F	16	Urban	Greater than 3	Living together	father	1
6	M	16	Urban	Less than or equal to 3	Living together	mother	1
7	M	16	Urban	Less than or equal to 3	Living together	mother	1
8	F	17	Urban	Greater than 3	Apart	mother	1
9	M	15	Urban	Less than or equal to 3	Apart	mother	1
10	M	15	Urban	Greater than 3	Living together	mother	1
11	F	15	Urban	Greater than 3	Living together	mother	1
12	F	15	Urban	Greater than 3	Living together	father	1
13	m	15	Urban	Less than or equal to 3	Living together	father	1
14	M	15	Urban	Greater than 3	Living together	mother	1
15	M	15	Urban	Greater than 3	Apart	other	1
16	F	16	Urban	Greater than 3	Living together	mother	1
17	F	16	Urban	Greater than 3	Living together	mother	1
18	F	16	Urban	Greater than 3	Living together	mother	1
19	M	17	Urban	Greater than 3	Living together	mother	1
20	M	16	Urban	Less than or equal to 3	Living together	father	1
21	M	15	Urban	Greater than 3	Living together	mother	1
22	M	15	Urban	Greater than 3	Living together	father	1
23	M	16	Urban	Less than or equal to 3	Living together	mother	1
24	M	16	Urban	Less than or equal to 3	Living together	mother	1
25	F	15	Rural	Greater than 3	Living together	mother	1
26	F	16	Urban	Greater than 3	Living together	mother	1

After (Clean)



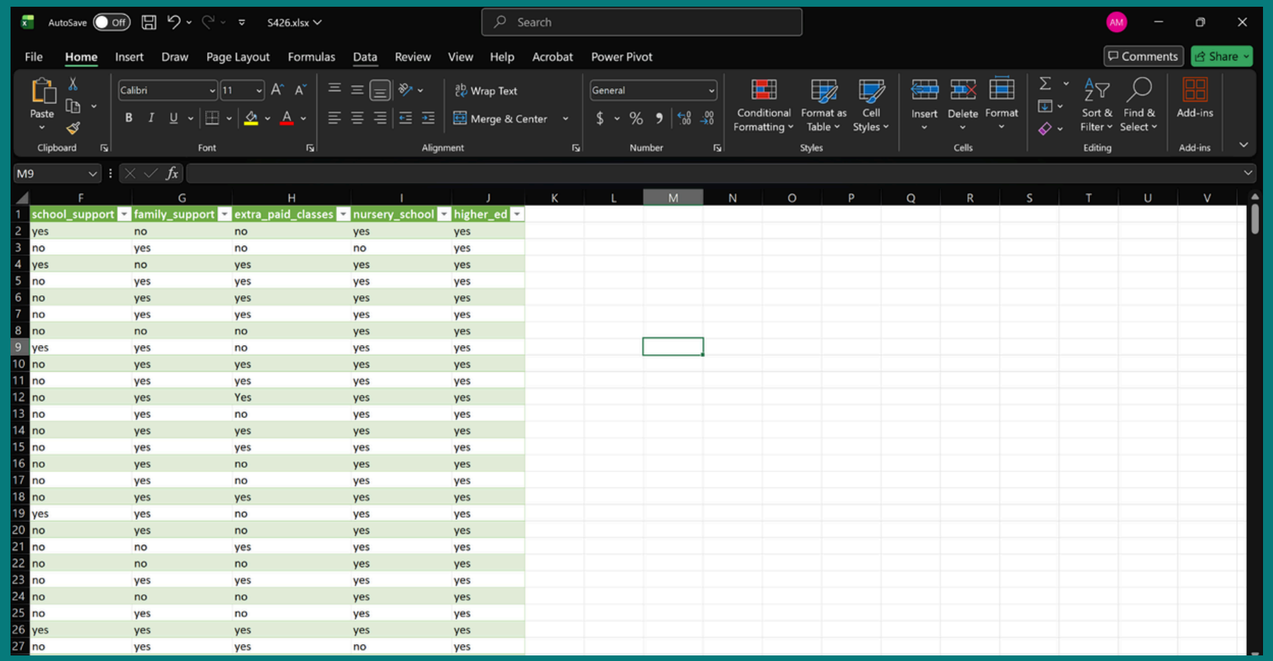
student_id	sex	age	address_type	family_size	parent_status	guardian	school_id	sex2
1	F	18	Urban	Greater than 3	Apart	mother	1	F
2	F	17	Urban	Greater than 3	Living together	father	1	F
3	F	15	Urban	Less than or equal to 3	Living together	mother	1	F
4	F	15	Urban	Greater than 3	Living together	mother	1	F
5	F	16	Urban	Greater than 3	Living together	father	1	F
6	M	16	Urban	Less than or equal to 3	Living together	mother	1	M
7	M	16	Urban	Less than or equal to 3	Living together	mother	1	M
8	F	17	Urban	Greater than 3	Apart	mother	1	F
9	M	15	Urban	Less than or equal to 3	Apart	mother	1	M
10	M	15	Urban	Greater than 3	Living together	mother	1	M
11	F	15	Urban	Greater than 3	Living together	mother	1	F
12	F	15	Urban	Greater than 3	Living together	father	1	F
13	m	15	Urban	Less than or equal to 3	Living together	father	1	M
14	M	15	Urban	Greater than 3	Living together	mother	1	M
15	M	15	Urban	Greater than 3	Apart	other	1	M
16	F	16	Urban	Greater than 3	Living together	mother	1	F
17	F	16	Urban	Greater than 3	Living together	mother	1	F
18	F	16	Urban	Greater than 3	Living together	mother	1	F
19	M	17	Urban	Greater than 3	Living together	mother	1	M
20	M	16	Urban	Less than or equal to 3	Living together	father	1	M
21	M	15	Urban	Greater than 3	Living together	mother	1	M
22	M	15	Urban	Greater than 3	Living together	father	1	M
23	M	16	Urban	Less than or equal to 3	Living together	mother	1	M
24	M	16	Urban	Less than or equal to 3	Living together	mother	1	M
25	F	15	Rural	Greater than 3	Living together	mother	1	F
26	F	16	Urban	Greater than 3	Living together	mother	1	F

Process : 1. Make Helper Column
2.[sex]=IF(UPPER(LEFT(TRIM(B2),1))="F","F","M")
3. Copy Paste New Values
4.Delete Helper Column

Columns Affected : sex

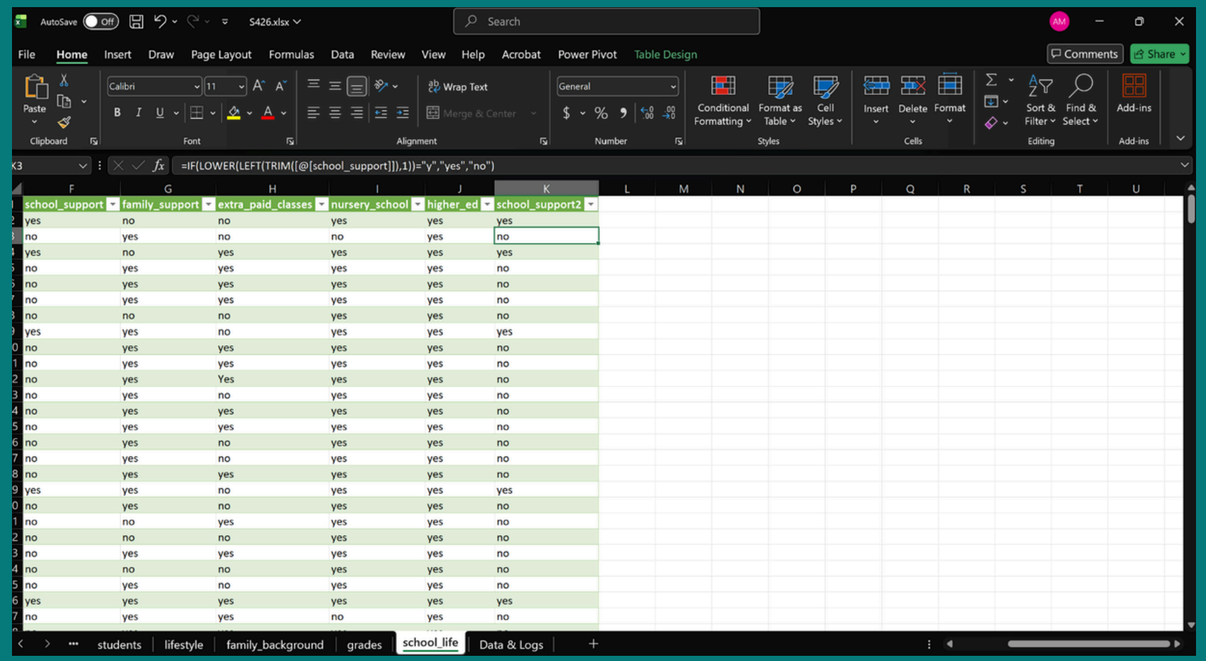
Standardize Yes/No

Before (Raw)



	school_support	family_support	extra_paid_classes	nursery_school	higher_ed
2	yes	no	no	yes	yes
3	no	yes	no	no	yes
4	yes	no	yes	yes	yes
5	no	yes	yes	yes	yes
6	no	yes	yes	yes	yes
7	no	yes	yes	yes	yes
8	no	no	no	yes	yes
9	yes	yes	no	yes	yes
10	no	yes	yes	yes	yes
11	no	yes	yes	yes	yes
12	no	yes	Yes	yes	yes
13	no	yes	no	yes	yes
14	no	yes	yes	yes	yes
15	no	yes	yes	yes	yes
16	no	yes	no	yes	yes
17	no	yes	no	yes	yes
18	no	yes	yes	yes	yes
19	yes	yes	no	yes	yes
20	no	yes	no	yes	yes
21	no	no	no	yes	yes
22	no	no	no	yes	yes
23	no	yes	yes	yes	yes
24	no	no	no	yes	yes
25	no	yes	no	yes	yes
26	yes	yes	yes	yes	yes
27	no	yes	yes	no	yes

After (Clean)



	school_support	family_support	extra_paid_classes	nursery_school	higher_ed	school_support2
2	yes	no	no	yes	yes	yes
3	no	yes	no	no	yes	no
4	yes	no	yes	yes	yes	yes
5	no	yes	yes	yes	yes	no
6	no	yes	yes	yes	yes	no
7	no	yes	yes	yes	yes	no
8	no	no	no	yes	yes	no
9	yes	yes	no	yes	yes	yes
10	no	yes	yes	yes	yes	no
11	no	yes	yes	yes	yes	no
12	no	yes	Yes	yes	yes	no
13	no	yes	no	yes	yes	no
14	no	yes	yes	yes	yes	no
15	no	yes	yes	yes	yes	no
16	no	yes	no	yes	yes	no
17	no	yes	no	yes	yes	no
18	no	yes	yes	yes	yes	no
19	yes	yes	no	yes	yes	yes
20	no	yes	no	yes	yes	no
21	no	no	yes	yes	yes	no
22	no	no	no	yes	yes	no
23	no	yes	yes	yes	yes	no
24	no	no	no	yes	yes	no
25	no	yes	no	yes	yes	no
26	yes	yes	yes	yes	yes	yes
27	no	yes	yes	no	yes	no

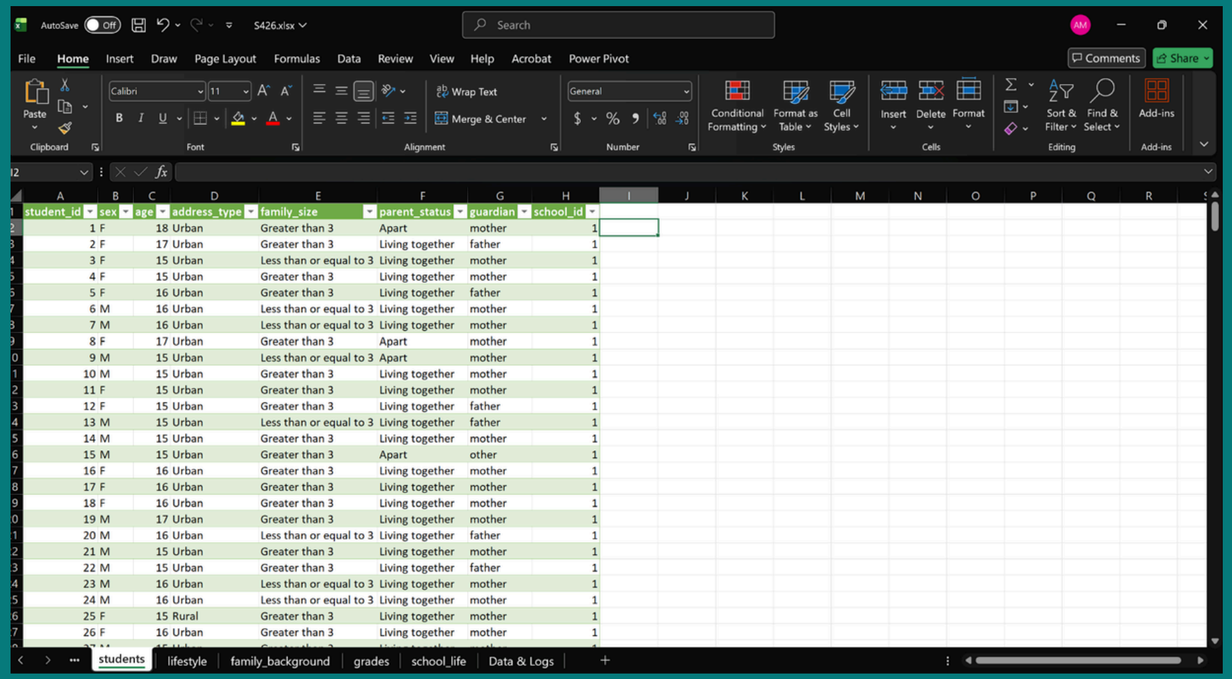
Process : 1.Make Helper Column
2.[ALL]=IF(LOWER(LEFT(TRIM(E2),1))="y","yes","no")
3.Copy Paste New Values
4.Delete Helper Column

Columns Affected :

- school_support(slide),
- family support,
- extra_paid_classes,
- activities,
- internat_access

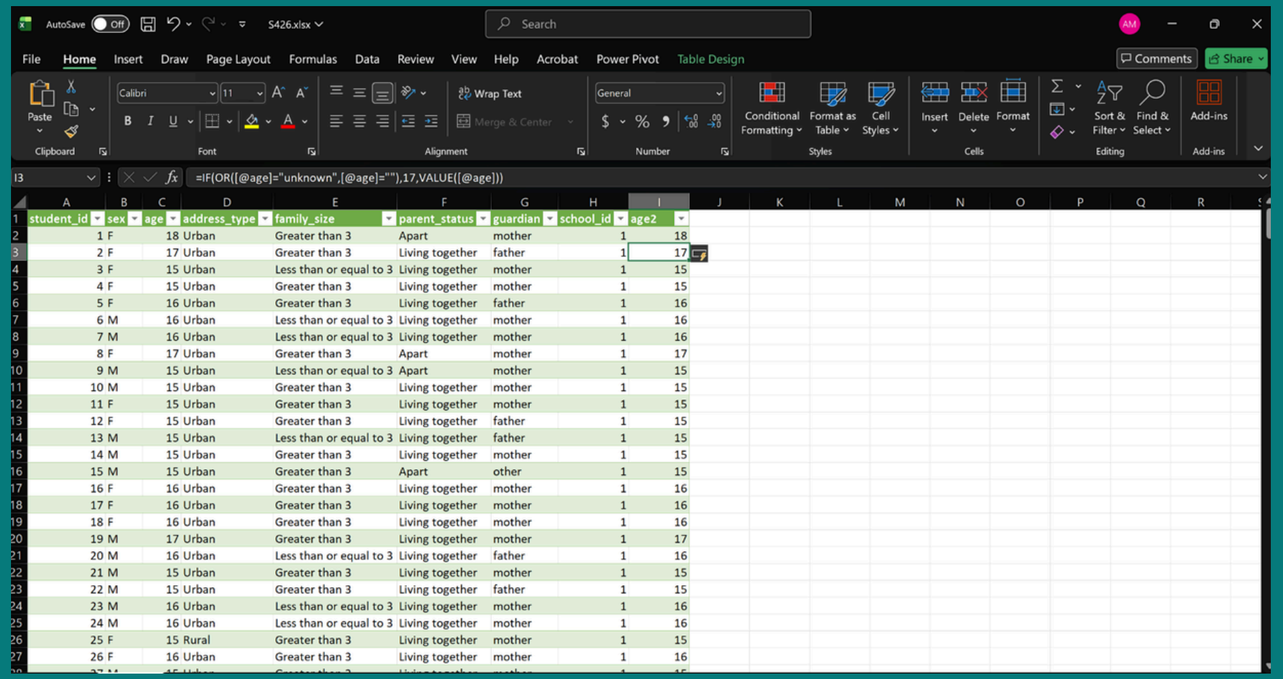
Fix Mixed Data Types

Before (Raw)



student_id	sex	age	address_type	family_size	parent_status	guardian	school_id
1	F	18	Urban	Greater than 3	Apart	mother	1
2	F	17	Urban	Greater than 3	Living together	father	1
3	F	15	Urban	Less than or equal to 3	Living together	mother	1
4	F	15	Urban	Greater than 3	Living together	mother	1
5	F	16	Urban	Greater than 3	Living together	father	1
6	M	16	Urban	Less than or equal to 3	Living together	mother	1
7	M	16	Urban	Less than or equal to 3	Living together	mother	1
8	F	17	Urban	Greater than 3	Apart	mother	1
9	M	15	Urban	Less than or equal to 3	Apart	mother	1
10	M	15	Urban	Greater than 3	Living together	mother	1
11	F	15	Urban	Greater than 3	Living together	mother	1
12	F	15	Urban	Greater than 3	Living together	father	1
13	M	15	Urban	Less than or equal to 3	Living together	father	1
14	M	15	Urban	Greater than 3	Living together	mother	1
15	M	15	Urban	Greater than 3	Apart	other	1
16	F	16	Urban	Greater than 3	Living together	mother	1
17	F	16	Urban	Greater than 3	Living together	mother	1
18	F	16	Urban	Greater than 3	Living together	mother	1
19	M	17	Urban	Greater than 3	Living together	mother	1
20	M	16	Urban	Less than or equal to 3	Living together	father	1
21	M	15	Urban	Greater than 3	Living together	mother	1
22	M	15	Urban	Greater than 3	Living together	father	1
23	M	16	Urban	Less than or equal to 3	Living together	mother	1
24	M	16	Urban	Less than or equal to 3	Living together	mother	1
25	F	15	Rural	Greater than 3	Living together	mother	1
26	F	16	Urban	Greater than 3	Living together	mother	1

After (Clean)



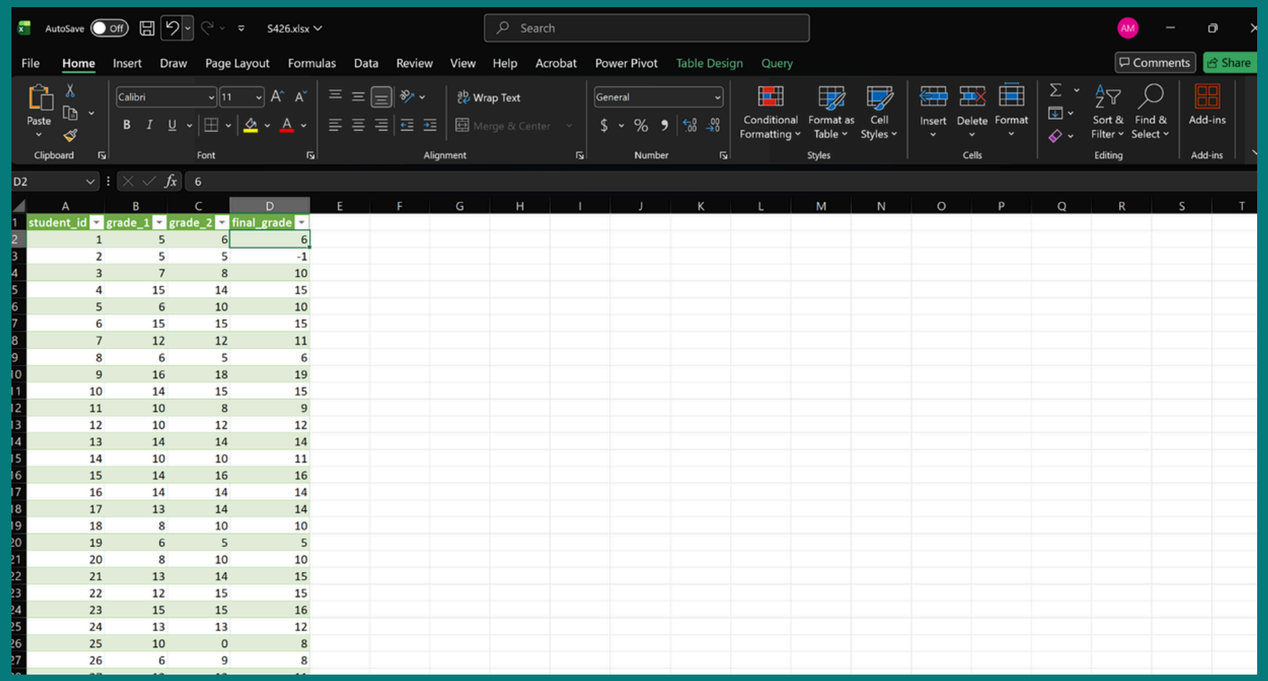
student_id	sex	age	address_type	family_size	parent_status	guardian	school_id	age2
1	F	18	Urban	Greater than 3	Apart	mother	1	18
2	F	17	Urban	Greater than 3	Living together	father	1	17
3	F	15	Urban	Less than or equal to 3	Living together	mother	1	15
4	F	15	Urban	Greater than 3	Living together	mother	1	15
5	F	16	Urban	Greater than 3	Living together	father	1	16
6	M	16	Urban	Less than or equal to 3	Living together	mother	1	16
7	M	16	Urban	Less than or equal to 3	Living together	mother	1	16
8	F	17	Urban	Greater than 3	Apart	mother	1	17
9	M	15	Urban	Less than or equal to 3	Apart	mother	1	15
10	M	15	Urban	Greater than 3	Living together	mother	1	15
11	F	15	Urban	Greater than 3	Living together	mother	1	15
12	F	15	Urban	Greater than 3	Living together	father	1	15
13	M	15	Urban	Less than or equal to 3	Living together	father	1	15
14	M	15	Urban	Greater than 3	Living together	mother	1	15
15	M	15	Urban	Greater than 3	Apart	other	1	15
16	F	16	Urban	Greater than 3	Living together	mother	1	16
17	F	16	Urban	Greater than 3	Living together	mother	1	16
18	F	16	Urban	Greater than 3	Living together	mother	1	16
19	M	17	Urban	Greater than 3	Living together	mother	1	17
20	M	16	Urban	Less than or equal to 3	Living together	father	1	16
21	M	15	Urban	Greater than 3	Living together	mother	1	15
22	M	15	Urban	Greater than 3	Living together	father	1	15
23	M	16	Urban	Less than or equal to 3	Living together	mother	1	16
24	M	16	Urban	Less than or equal to 3	Living together	mother	1	16
25	F	15	Rural	Greater than 3	Living together	mother	1	15
26	F	16	Urban	Greater than 3	Living together	mother	1	16

Process : 1.Make Helper Column
2.=IF(OR(G2="unknown",G2=""),17,VALUE(G2))
3.Copy Paste New Values
4.Delete Helper Column

Columns Affected : age

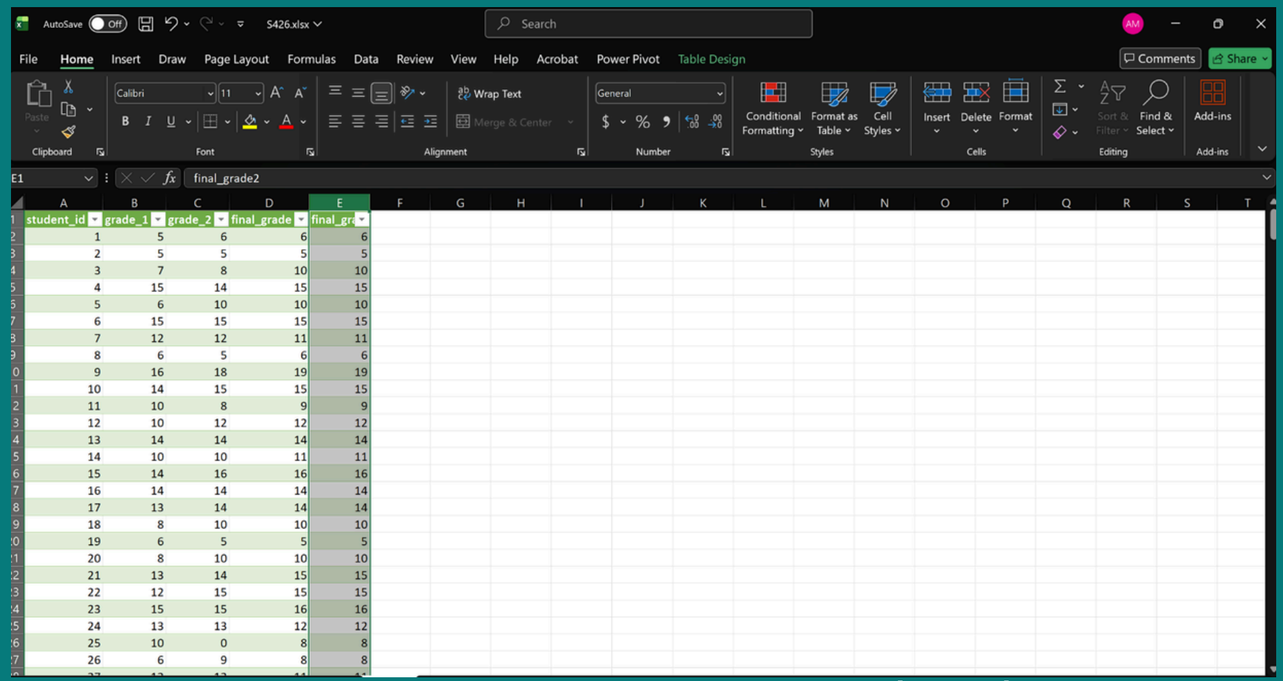
Fix Out of Range Values

Before (Raw)



student_id	grade_1	grade_2	final_grade
1	5	6	6
2	5	5	-1
3	7	8	10
4	15	14	15
5	6	10	10
6	15	15	15
7	12	12	11
8	6	5	6
9	16	18	19
10	14	15	15
11	10	8	9
12	10	12	12
13	14	14	14
14	10	10	11
15	14	16	16
16	14	14	14
17	13	14	14
18	8	10	10
19	6	5	5
20	8	10	10
21	13	14	15
22	12	15	15
23	15	15	16
24	13	13	12
25	10	0	8
26	6	9	8

After (Clean)



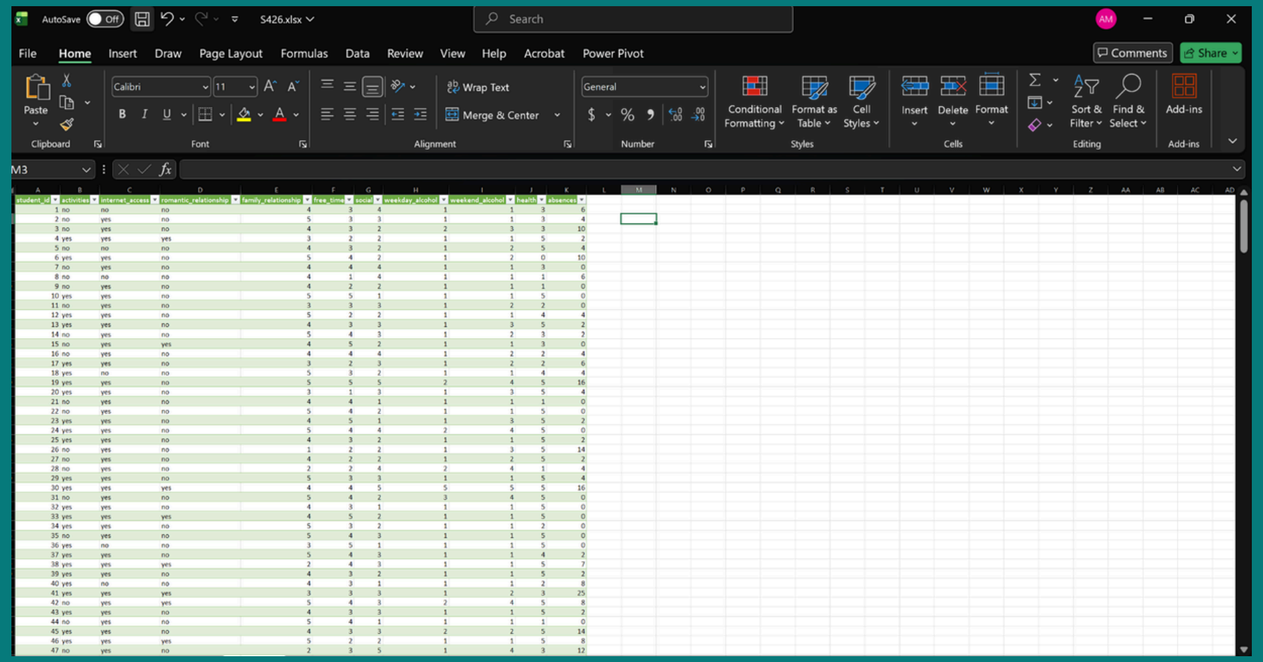
student_id	grade_1	grade_2	final_grade	final_gr
1	5	6	6	6
2	5	5	5	5
3	7	8	10	10
4	15	14	15	15
5	6	10	10	10
6	15	15	15	15
7	12	12	11	11
8	6	5	6	6
9	16	18	19	19
10	14	15	15	15
11	10	8	9	9
12	10	12	12	12
13	14	14	14	14
14	10	10	11	11
15	14	16	16	16
16	14	14	14	14
17	13	14	14	14
18	8	10	10	10
19	6	5	5	5
20	8	10	10	10
21	13	14	15	15
22	12	15	15	15
23	15	15	16	16
24	13	13	12	12
25	10	0	8	8
26	6	9	8	8

Process : 1.Make Helper Column
2.[grade]=IF(OR(D2<0,D2>20),AVERAGE(B2,C2),D2)
3.[absences]=IF(K2<0,0,K2)
4.Copy Paste New Values
5.Delete Helper Column

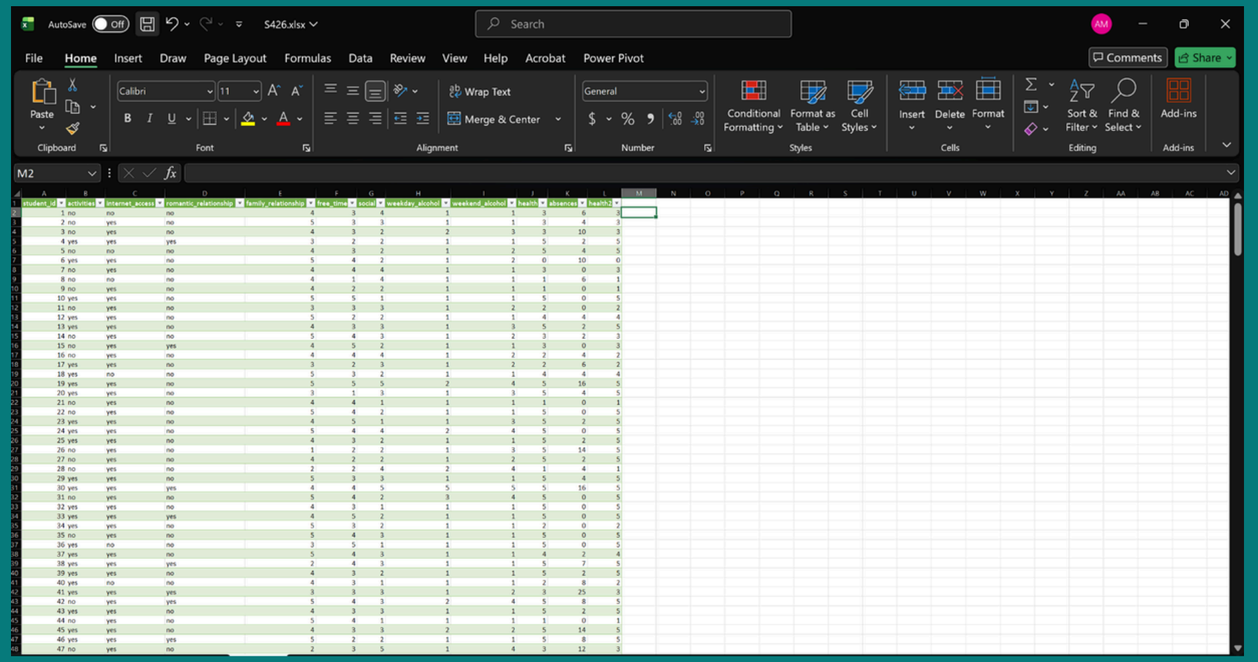
Columns Affected :
- final_grade(Slide),
- absences

Impute Missing Values

Before (Raw)



After (Clean)



Process :

1. Make Helper Column
2. [health]=MEDIAN(lifestyle!J2:J396)
3. [health]=IF(ISBLANK(J2),4,J2)
4. [absences]=IF(ISBLANK(K2),0,K2)
5. [grade_1]=IF(ISBLANK(B2),AVERAGE(C2,D2),B2)
6. [grade_2]=IF(ISBLANK(C2),AVERAGE(B2,D2),C2)
7. [final_grade]=IF(ISBLANK(D2),AVERAGE(B2,C2),D2)
8. Delete Helper Column

Columns Affected :

- age(already),
- health,
- absences,
- grade_1,
- grade_2,
- final_grade

Named Ranges and Tables

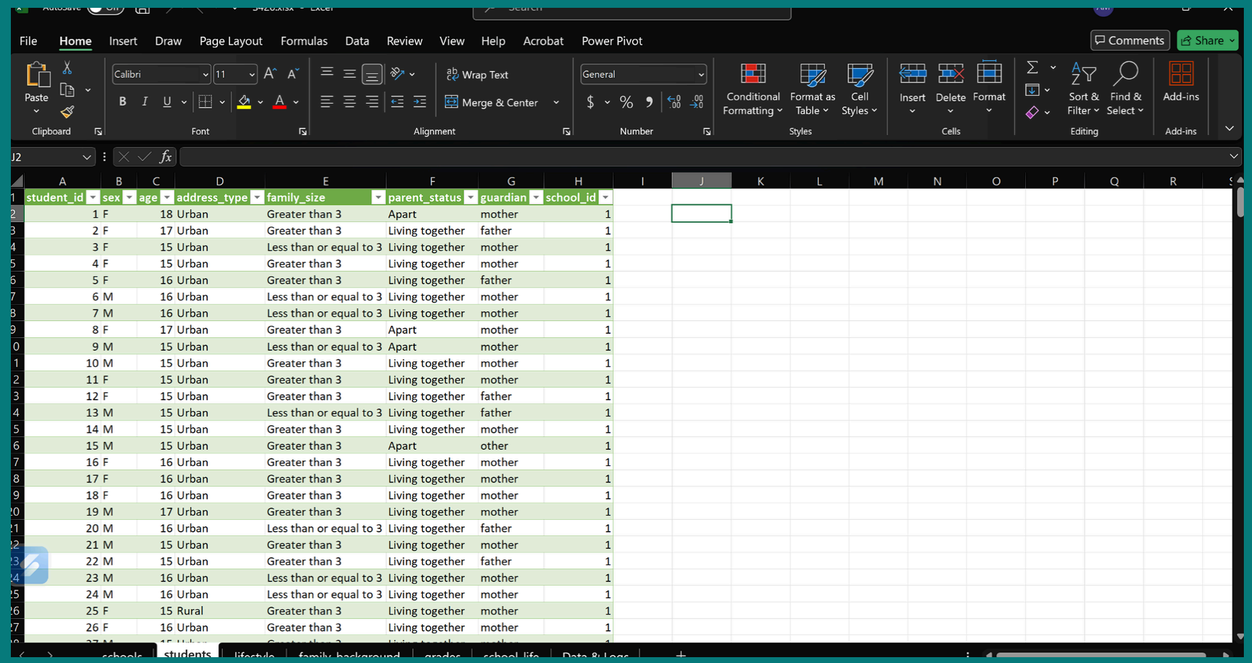
Standard Guideline Requirement
Mengubah dataset utama menjadi Excel Table
Membuat minimal 2 Named Ranges
Harus digunakan dalam pengolahan atau analisis.

Table Design Directory	
Students	tbl_Students
Grades	tbl_Grades
Lifestyle	tbl_Lifestyle
Family_background	tbl_FamilyBackground
School_life	tbl_SchoolLife
schools	tbl_Schools

Analysis Named Ranges Directory	
Average Final Grade	=AVERAGE(FinalGrades)
Highest Grade	=MAX(FinalGrades)
Lowest Grade	=MIN(FinalGrades)
Students Passing (>10)	=COUNTIF(FinalGrades,">10")
Pass Rate %	=COUNTIF(FinalGrades,">10")/COUNT(Final Grades)
Average Absences	=AVERAGE(StudentAbsences)
Students with 0 Absences	=COUNTIF(StudentAbsences,0)
High Absentees (>20 days)	=COUNTIF(StudentAbsences,">20")

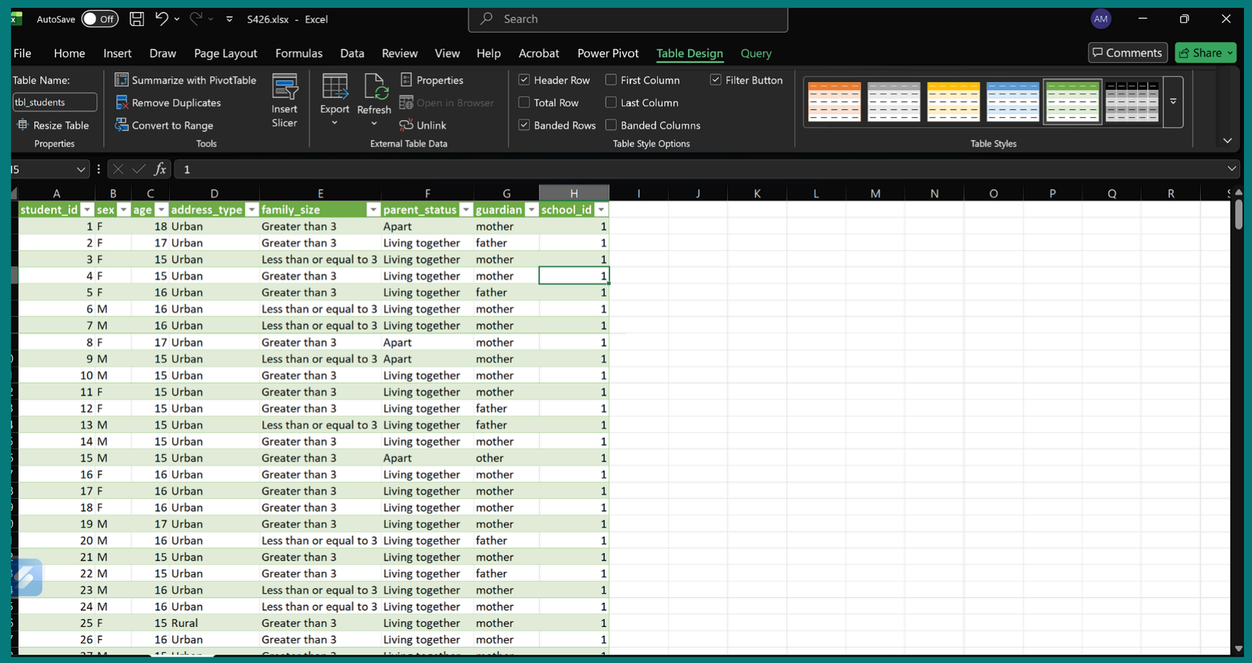
Table Design Order

Before (Raw)



student_id	sex	age	address_type	family_size	parent_status	guardian	school_id
1	F	18	Urban	Greater than 3	Apart	mother	1
2	F	17	Urban	Greater than 3	Living together	father	1
3	F	15	Urban	Less than or equal to 3	Living together	mother	1
4	F	15	Urban	Greater than 3	Living together	mother	1
5	F	16	Urban	Greater than 3	Living together	father	1
6	M	16	Urban	Less than or equal to 3	Living together	mother	1
7	M	16	Urban	Less than or equal to 3	Living together	mother	1
8	F	17	Urban	Greater than 3	Apart	mother	1
9	M	15	Urban	Less than or equal to 3	Apart	mother	1
10	M	15	Urban	Greater than 3	Living together	mother	1
11	F	15	Urban	Greater than 3	Living together	mother	1
12	F	15	Urban	Greater than 3	Living together	father	1
13	M	15	Urban	Less than or equal to 3	Living together	father	1
14	M	15	Urban	Greater than 3	Living together	mother	1
15	M	15	Urban	Greater than 3	Apart	other	1
16	F	16	Urban	Greater than 3	Living together	mother	1
17	F	16	Urban	Greater than 3	Living together	mother	1
18	F	16	Urban	Greater than 3	Living together	mother	1
19	M	17	Urban	Greater than 3	Living together	mother	1
20	M	16	Urban	Less than or equal to 3	Living together	father	1
21	M	15	Urban	Greater than 3	Living together	mother	1
22	M	15	Urban	Greater than 3	Living together	father	1
23	M	16	Urban	Less than or equal to 3	Living together	mother	1
24	M	16	Urban	Less than or equal to 3	Living together	mother	1
25	F	15	Rural	Greater than 3	Living together	mother	1
26	F	16	Urban	Greater than 3	Living together	mother	1

After (Clean)



student_id	sex	age	address_type	family_size	parent_status	guardian	school_id
1	F	18	Urban	Greater than 3	Apart	mother	1
2	F	17	Urban	Greater than 3	Living together	father	1
3	F	15	Urban	Less than or equal to 3	Living together	mother	1
4	F	15	Urban	Greater than 3	Living together	mother	1
5	F	16	Urban	Greater than 3	Living together	father	1
6	M	16	Urban	Less than or equal to 3	Living together	mother	1
7	M	16	Urban	Less than or equal to 3	Living together	mother	1
8	F	17	Urban	Greater than 3	Apart	mother	1
9	M	15	Urban	Less than or equal to 3	Apart	mother	1
10	M	15	Urban	Greater than 3	Living together	mother	1
11	F	15	Urban	Greater than 3	Living together	mother	1
12	F	15	Urban	Greater than 3	Living together	father	1
13	M	15	Urban	Less than or equal to 3	Living together	father	1
14	M	15	Urban	Greater than 3	Living together	mother	1
15	M	15	Urban	Greater than 3	Apart	other	1
16	F	16	Urban	Greater than 3	Living together	mother	1
17	F	16	Urban	Greater than 3	Living together	mother	1
18	F	16	Urban	Greater than 3	Living together	mother	1
19	M	17	Urban	Greater than 3	Living together	mother	1
20	M	16	Urban	Less than or equal to 3	Living together	father	1
21	M	15	Urban	Greater than 3	Living together	mother	1
22	M	15	Urban	Greater than 3	Living together	father	1
23	M	16	Urban	Less than or equal to 3	Living together	mother	1
24	M	16	Urban	Less than or equal to 3	Living together	mother	1
25	F	15	Rural	Greater than 3	Living together	mother	1
26	F	16	Urban	Greater than 3	Living together	mother	1

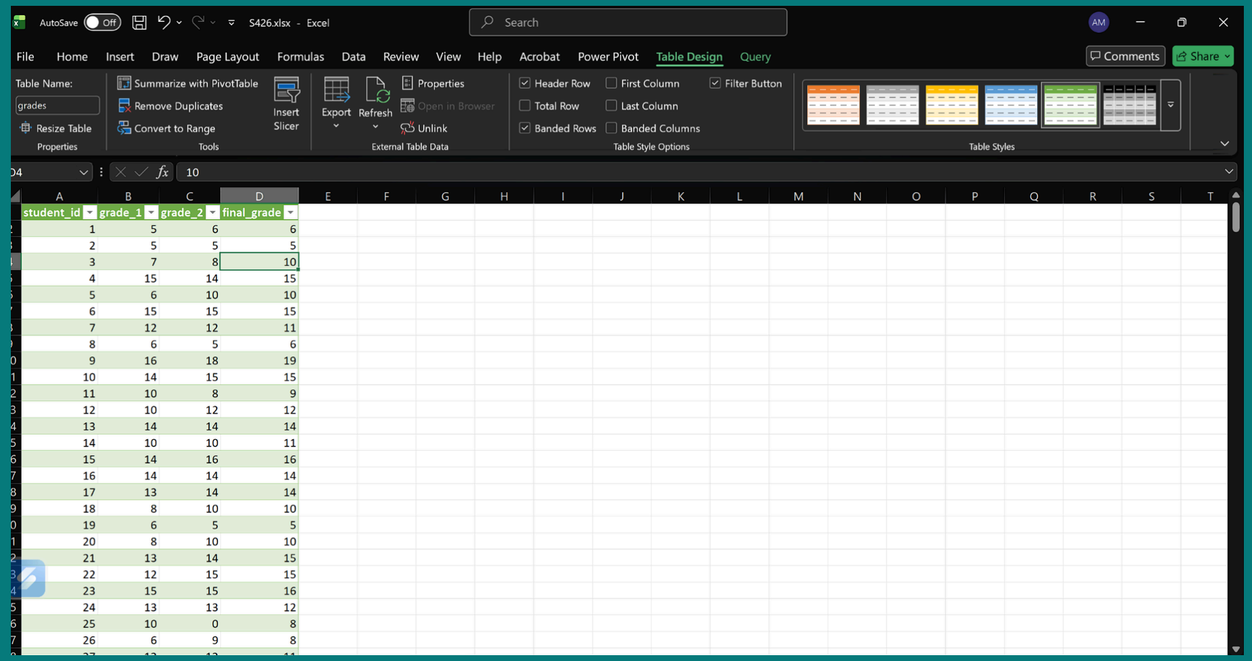
Process : Mengubah dataset utama menjadi Excel Table

Table Affected :

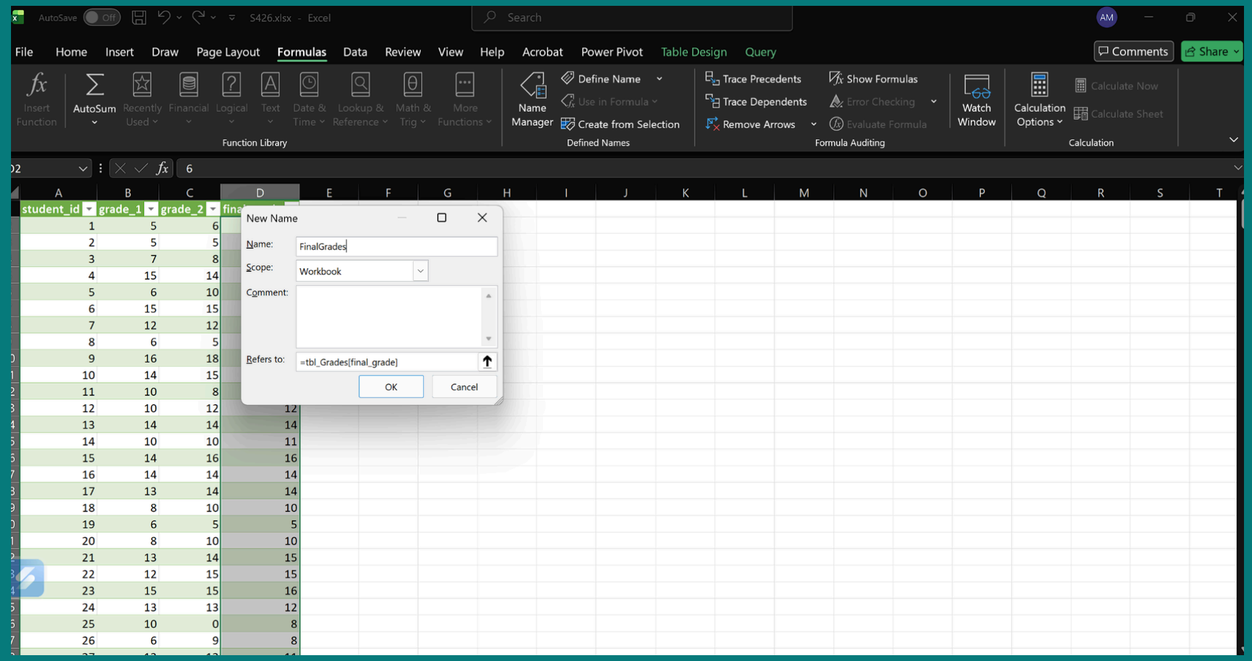
Table Design Directory	
Students	tbl_Students
Grades	tbl_Grades
Lifestyle	tbl_Lifestyle
Family_background	tbl_FamilyBackground
School_life	tbl_SchoolLife
schools	tbl_Schools

Create Named Ranges

Before (Raw)



After (Clean)



Process :

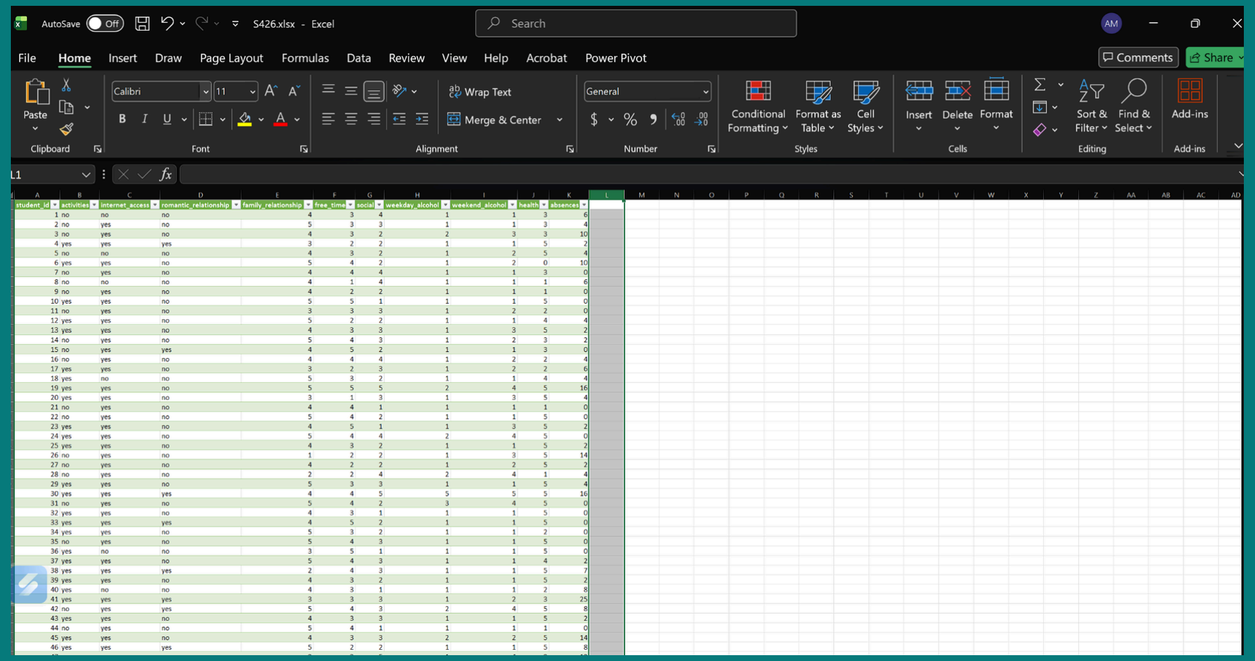
```
=AVERAGE(FinalGrades)
=MAX(FinalGrades)
=MIN(FinalGrades)
=COUNTIF(FinalGrades,">10")
=COUNTIF(FinalGrades,">10")/COUNT(FinalGrades)
```

Columns Affected :

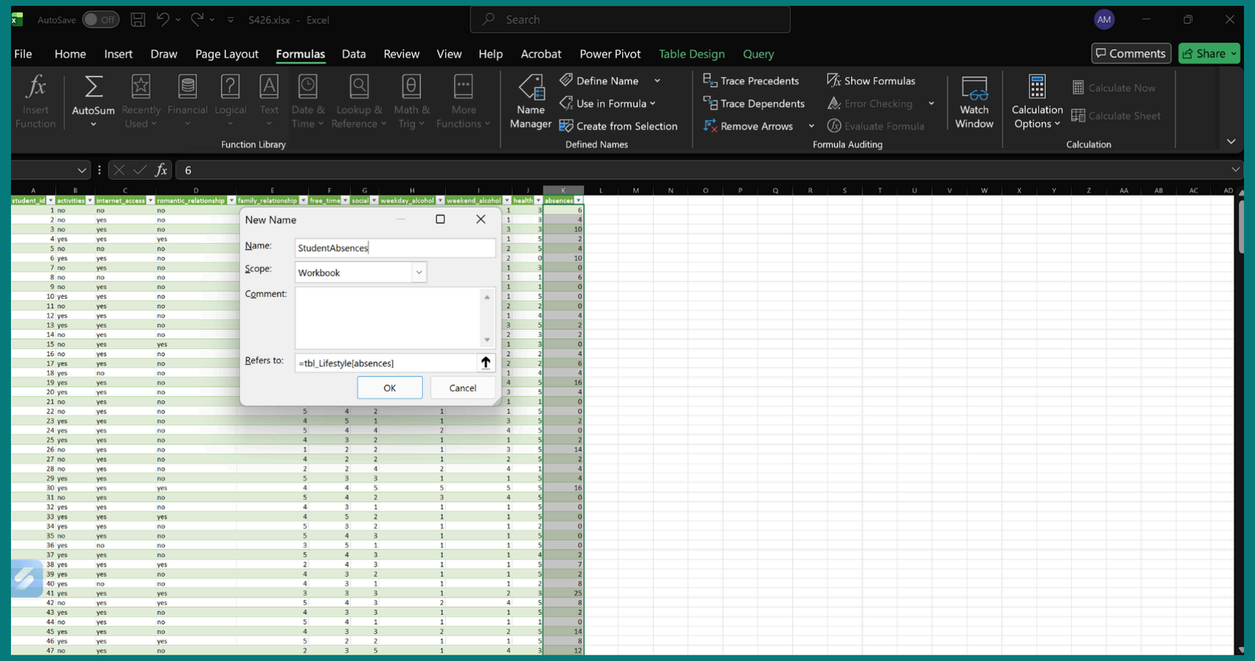
- final_grade

Create Named Ranges

Before (Raw)



After (Clean)



Process :

```
=AVERAGE(StudentAbsences)
=COUNTIF(StudentAbsences,0)
=COUNTIF(StudentAbsences,">20")
```

Columns Affected :

- absences

Analysis of Named Ranges

Analysis Named Ranges Directory	Result
Average Final Grade	10.21
Highest Grade	20
Lowest Grade	0
Students Passing (>10)	205
Pass Rate %	51.90%
Average Absences	6
Students with 0 Absences	120
High Absentees (>20 days)	14

Average Final Grade: 10.21 = the cohort average sits just above the passing threshold of 10 on the 0–20 Portuguese scale, meaning the average student is barely passing

Highest Grade: 20 = at least one student achieved a perfect score

Lowest Grade: 0 = at least one student scored zero, indicating complete non-performance or non-attendance at the final assessment

Students Passing (>10): 205 out of 395 = just over half the cohort is passing

Pass Rate: 51.90% = nearly half the student population is failing, which is a significant finding for a school context

Average Absences: 6 = on average students miss 6 school days, which is relatively low

Students with 0 Absences: 120 = almost a third of the cohort has perfect attendance

High Absentees (>20 days): 14 = a small but notable group of students with chronic

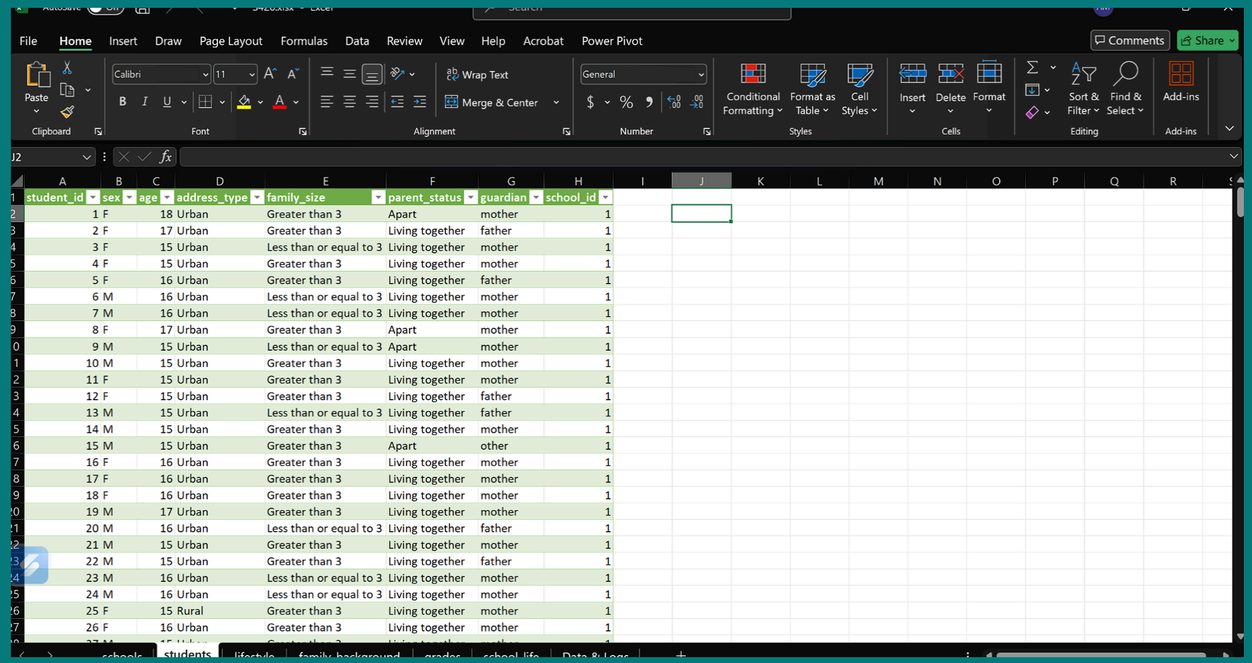
Logical, Lookup, and Conditional Functions

Standard Guideline Requirement. Pick 2 out of 3.	
Logical Functions	Chosen
Lookup Functions	Chosen
Conditional Functions	N/A

Formula Functions Directory	
Logical Functions	=IF(tbl_Grades[@final_grade]>=16,"Excellent",IF(tbl_Grades[@final_grade]>=12,"Good",IF(tbl_Grades[@final_grade]>=8,"Average","Below Average")))
	=IF(AND(tbl_Lifestyle[absences]>15,tbl_Grades[final_grade]<10),"At Risk","OK")
Lookup Functions (XLOOKUP)	=XLOOKUP([@[school_id]],tbl_Schools[school_id],tbl_Schools[school_name],"Unknown")
Conditional Functions	N/A

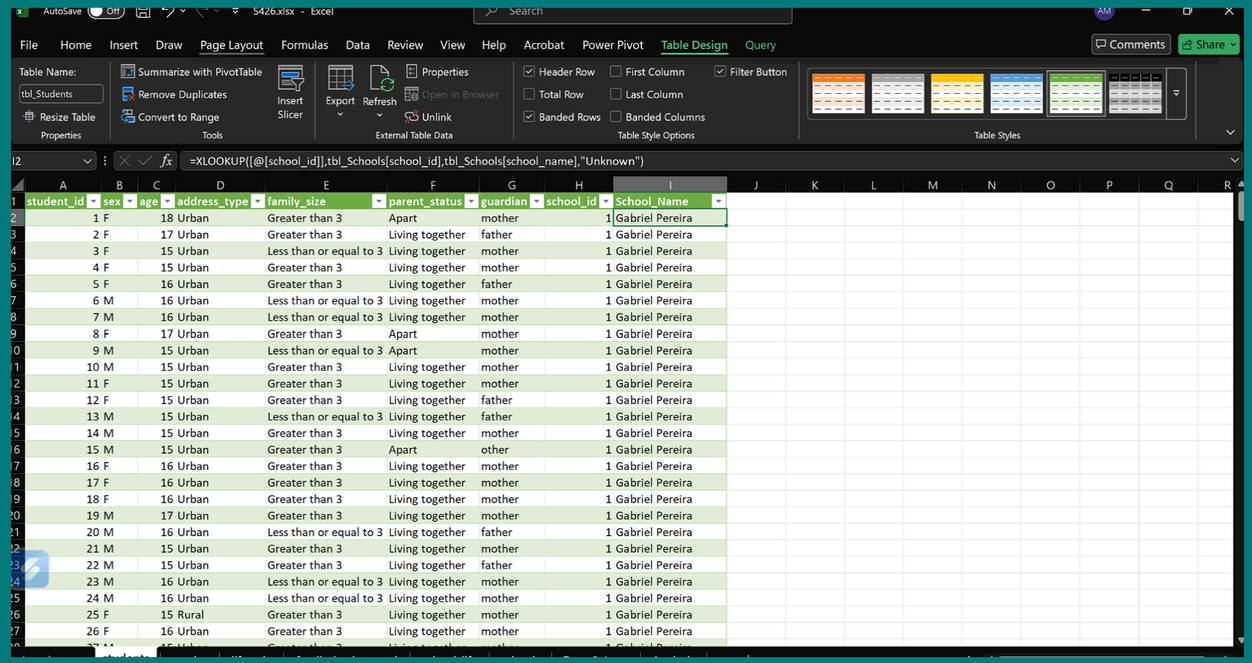
Lookup Functions (XLOOKUP)

Before (Raw)



student_id	sex	age	address_type	family_size	parent_status	guardian	school_id
1	F	18	Urban	Greater than 3	Apart	mother	1
2	F	17	Urban	Greater than 3	Living together	father	1
3	F	15	Urban	Less than or equal to 3	Living together	mother	1
4	F	15	Urban	Greater than 3	Living together	mother	1
5	F	16	Urban	Greater than 3	Living together	father	1
6	M	16	Urban	Less than or equal to 3	Living together	mother	1
7	M	16	Urban	Less than or equal to 3	Living together	mother	1
8	F	17	Urban	Greater than 3	Apart	mother	1
9	M	15	Urban	Less than or equal to 3	Apart	mother	1
10	M	15	Urban	Greater than 3	Living together	mother	1
11	F	15	Urban	Greater than 3	Living together	mother	1
12	F	15	Urban	Greater than 3	Living together	father	1
13	M	15	Urban	Less than or equal to 3	Living together	father	1
14	M	15	Urban	Greater than 3	Living together	mother	1
15	M	15	Urban	Greater than 3	Apart	other	1
16	F	16	Urban	Greater than 3	Living together	mother	1
17	F	16	Urban	Greater than 3	Living together	mother	1
18	F	16	Urban	Greater than 3	Living together	mother	1
19	M	17	Urban	Greater than 3	Living together	mother	1
20	M	16	Urban	Less than or equal to 3	Living together	father	1
21	M	15	Urban	Greater than 3	Living together	mother	1
22	M	15	Urban	Greater than 3	Living together	father	1
23	M	16	Urban	Less than or equal to 3	Living together	mother	1
24	M	16	Urban	Less than or equal to 3	Living together	mother	1
25	F	15	Rural	Greater than 3	Living together	mother	1
26	F	16	Urban	Greater than 3	Living together	mother	1

After (Clean)

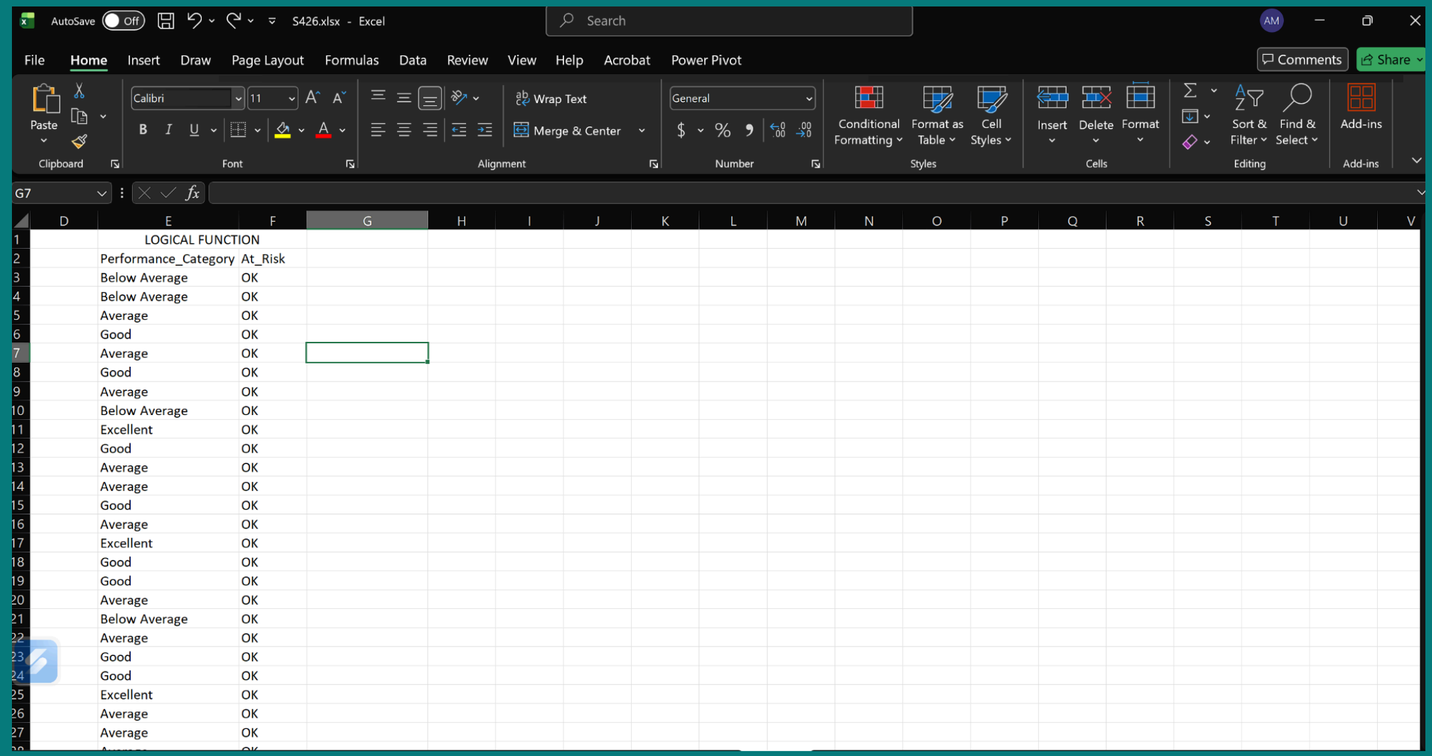


student_id	sex	age	address_type	family_size	parent_status	guardian	school_id	School_Name
1	F	18	Urban	Greater than 3	Apart	mother	1	Gabriel Pereira
2	F	17	Urban	Greater than 3	Living together	father	1	Gabriel Pereira
3	F	15	Urban	Less than or equal to 3	Living together	mother	1	Gabriel Pereira
4	F	15	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
5	F	16	Urban	Greater than 3	Living together	father	1	Gabriel Pereira
6	M	16	Urban	Less than or equal to 3	Living together	mother	1	Gabriel Pereira
7	M	16	Urban	Less than or equal to 3	Living together	mother	1	Gabriel Pereira
8	F	17	Urban	Greater than 3	Apart	mother	1	Gabriel Pereira
9	M	15	Urban	Less than or equal to 3	Apart	mother	1	Gabriel Pereira
10	M	15	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
11	F	15	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
12	F	15	Urban	Greater than 3	Living together	father	1	Gabriel Pereira
13	M	15	Urban	Less than or equal to 3	Living together	father	1	Gabriel Pereira
14	M	15	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
15	M	15	Urban	Greater than 3	Apart	other	1	Gabriel Pereira
16	F	16	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
17	F	16	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
18	F	16	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
19	M	17	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
20	M	16	Urban	Less than or equal to 3	Living together	father	1	Gabriel Pereira
21	M	15	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira
22	M	15	Urban	Greater than 3	Living together	father	1	Gabriel Pereira
23	M	16	Urban	Less than or equal to 3	Living together	mother	1	Gabriel Pereira
24	M	16	Urban	Less than or equal to 3	Living together	mother	1	Gabriel Pereira
25	F	15	Rural	Greater than 3	Living together	mother	1	Gabriel Pereira
26	F	16	Urban	Greater than 3	Living together	mother	1	Gabriel Pereira

Process : =XLOOKUP([@[school_id]],tbl_Schools[school_id],tbl_Schools[school_name],"Unknown")

Context: The Before/After comparison confirms XLOOKUP successfully enriched the student table with human-readable school names (Gabriel Pereira, Mousinho da Silveira) from their numeric IDs. This is important for dashboard readability charts and slicers now display actual school names rather than meaningless ID numbers 1 and 2.

Logical Functions



	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1		LOGICAL FUNCTION																	
2		Performance_Category	At_Risk																
3		Below Average	OK																
4		Below Average	OK																
5		Average	OK																
6		Good	OK																
7		Average	OK																
8		Good	OK																
9		Average	OK																
10		Below Average	OK																
11		Excellent	OK																
12		Good	OK																
13		Average	OK																
14		Average	OK																
15		Good	OK																
16		Average	OK																
17		Excellent	OK																
18		Good	OK																
19		Good	OK																
20		Average	OK																
21		Below Average	OK																
22		Average	OK																
23		Good	OK																
24		Good	OK																
25		Excellent	OK																
26		Average	OK																
27		Average	OK																

Process :

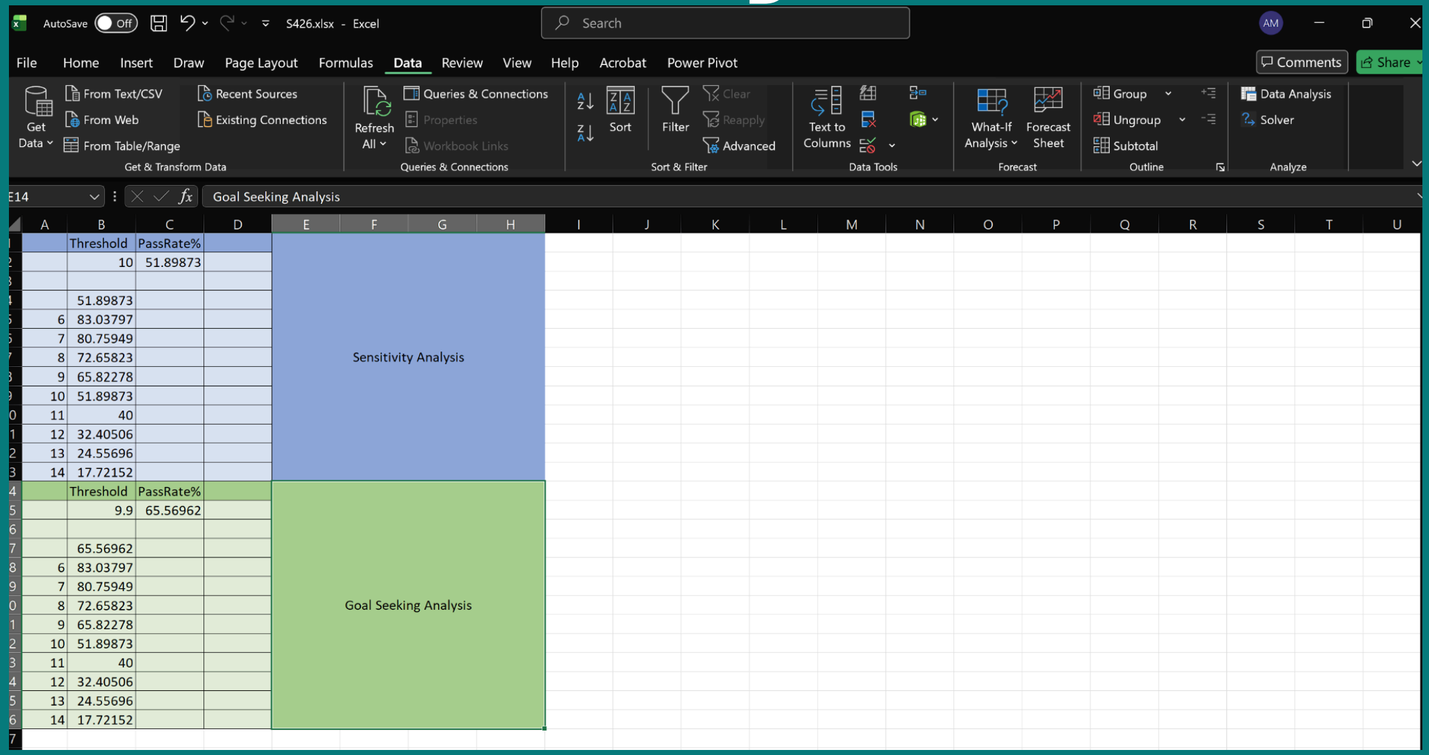
```
=IF(tbl_Grades[@final_grade]>=16,"Excellent",IF(tbl_Grades[@final_grade]>=12,"Good",IF(tbl_Grades[@final_grade]>=8,"Average","Below Average")))
=IF(AND(tbl_Lifestyle[absences]>15,tbl_Grades[final_grade]<10),"At Risk","OK")
```

Context: Performance Category With At_Risk =The output shows the majority of students falling into "Below Average", "Average", and "Good" categories with very few "Excellent" entries visible in the first rows. The At_Risk column showing predominantly "OK" in the early rows suggests the worst-performing students are concentrated further down the dataset. This classification system gives teachers an immediate actionable view, they can filter for "At Risk" students and intervene before final assessments.

Analysis with Data Tables

Standard Guideline Requirement	
Sensitivity Analysis	Chosen
What If Analysis	N/A
Goal Seek Analysis	Chosen

Analysis



	Threshold	PassRate%
	10	51.89873
	51.89873	
6	83.03797	
7	80.75949	
8	72.65823	
9	65.82278	
10	51.89873	
11	40	
12	32.40506	
13	24.55696	
14	17.72152	

	Threshold	PassRate%
	9.9	65.56962
	65.56962	
6	83.03797	
7	80.75949	
8	72.65823	
9	65.82278	
10	51.89873	
11	40	
12	32.40506	
13	24.55696	
14	17.72152	

Process :

A Sensitivity Analysis and Goal Seek Analysis were performed to model how changes in the passing grade threshold affect the overall student pass rate. This is analytically relevant for school policy understanding what threshold produces what outcome.

Sensitivity Analysis :

- Base formula in C2: =COUNTIF(FinalGrades,">"&B2)/COUNT(FinalGrades)*100 with threshold variable in B2 set to 10.
- Threshold values 6 through 14 entered in column A (rows 5–13).
- Selected range A4:C13 → Data → What-If Analysis → Data Table → Column input cell: B2 → OK.

Goal Seek Analysis :

- Posed the question: "What passing threshold is needed to achieve a 65% pass rate?"
- Data → What-If Analysis → Goal Seek → Set cell: C15 → To value: 65 → By changing cell: B15 → OK.

Context:

Sensitivity Analysis (blue section): The table reveals a steep non-linear relationship between the passing threshold and pass rate:

- At threshold 6: 83.04% of students pass — nearly the whole cohort
- At threshold 10 (current): 51.89% pass — just over half
- At threshold 14: 17.72% pass — almost everyone fails

This tells school administrators something important: the current passing threshold of 10 (half of the 20-point maximum) is already quite demanding for this cohort. Raising it even slightly to 11 drops the pass rate to 40%, meaning roughly 50 more students would fail. This is a genuine policy-relevant insight, not just a formula exercise

Goal Seek Analysis (green section): The question posed was: what threshold is needed to achieve a 65% pass rate?

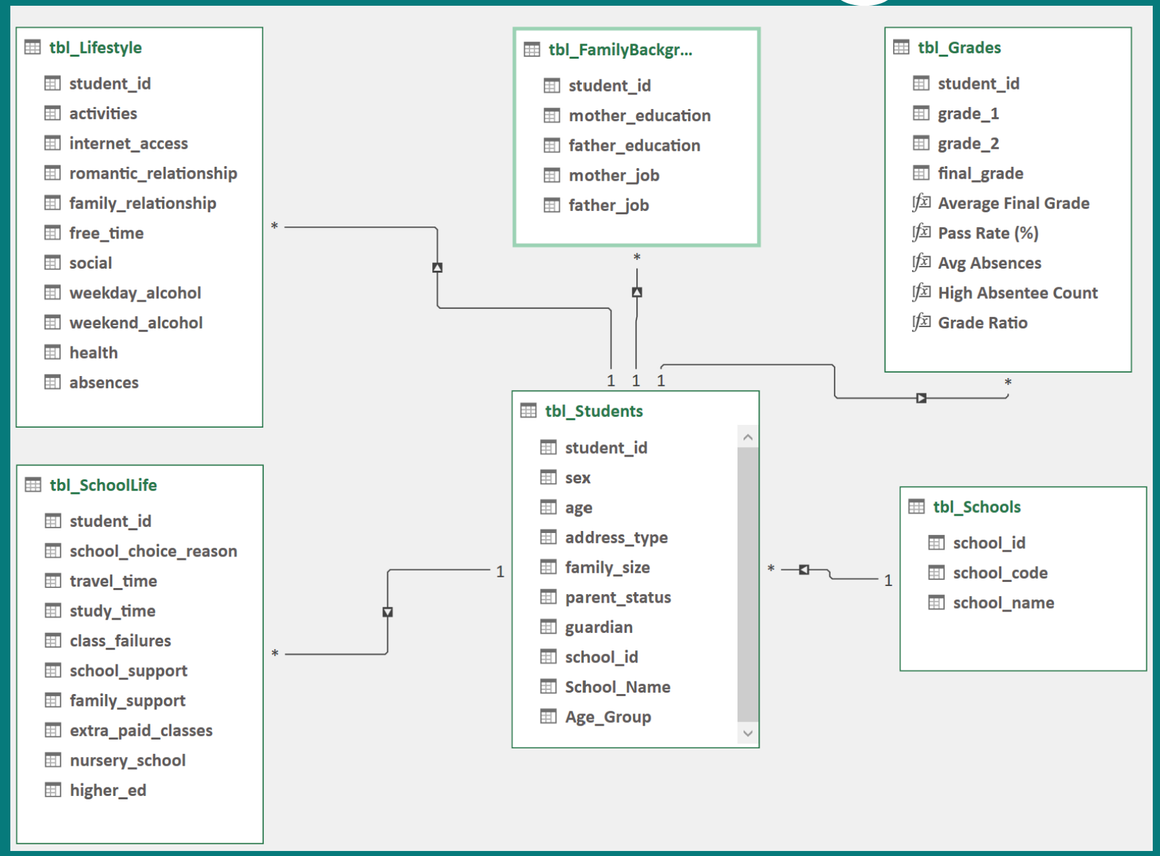
The answer is 9.9 meaning the passing grade would need to be lowered by just 0.1 points from the current 10 to push 13% more students into passing. This is a meaningful finding a very small policy adjustment has a significant impact on student outcomes, which reflects the fact that many students are clustered just below the passing line.

Power Pivot and Data Model

Standard Guideline Requirement	
Membuat Relationship dalam Data Model	Mandatory
Minimal 2 Tabel dalam Data Model	Mandatory
Minimal 1 Relationship aktif	Mandatory
Minimal 2 Calculated fields / measures	Mandatory
Minimal 1 Cube Functions	Mandatory

Formula Directory	
Calculated Field/Measures	Average Final Grade: =AVERAGEX(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade]))
	Pass Rate (%): =DIVIDE(COUNTROWS(FILTER(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade])>10)), COUNTROWS(tbl_Students))
	Avg Absences: =AVERAGE(tbl_Lifestyle[absences])
	High Absentee Count: =COUNTROWS(FILTER(tbl_Lifestyle,tbl_Lifestyle[absenc es]>20))
Cube Functions	Average Final Grade: =CUBEVALUE("ThisWorkbookDataModel","[Measures].[A verage Final Grade]")
	Pass Rate (%): =CUBEVALUE("ThisWorkbookDataModel","[Measures].[P ass Rate (%)]")
	Average Absences: =CUBEVALUE("ThisWorkbookDataModel","[Measures].[A vg Absences]")
	High Absentee Count: =CUBEVALUE("ThisWorkbookDataModel","[Measures].[H gh Absentee Count]")

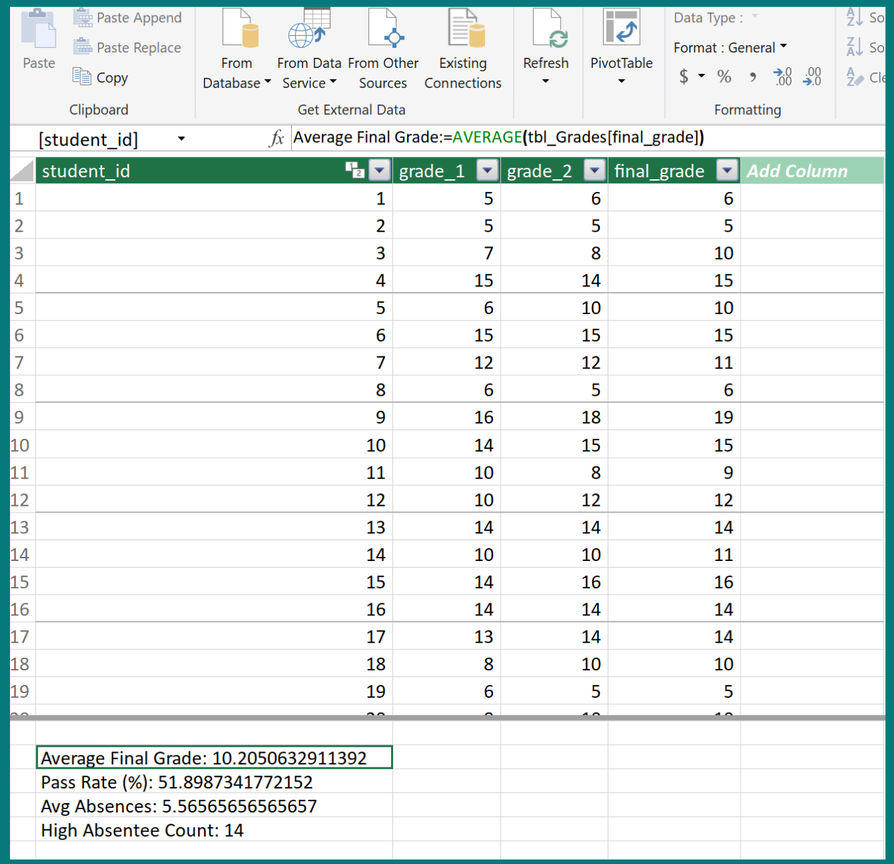
Data Model Diagram View



Process : tbl_Schools[school_id] → drag to → tbl_Students[school_id]
tbl_Students[student_id] → drag to → tbl_Grades[student_id]
tbl_Students[student_id] → drag to → tbl_Lifestyle[student_id]
tbl_Students[student_id] → drag to → tbl_FamilyBackground[student_id]
tbl_Students[student_id] → drag to → tbl_SchoolLife[student_id]

Context: The star schema with tbl_Students at the center reflects the real-world structure of school data, every piece of information (grades, lifestyle, family background, school life) belongs to a student. The 5 active relationships confirm the model is correctly built. The presence of School_Name and Age_Group as calculated columns directly on tbl_Students shows deliberate design thinking, bringing frequently-needed attributes close to the central table to avoid multi-hop filter problems encountered during development.

Calculated Field & Measures



	student_id	grade_1	grade_2	final_grade	Add Column
1	1	5	6	6	
2	2	5	5	5	
3	3	7	8	10	
4	4	15	14	15	
5	5	6	10	10	
6	6	15	15	15	
7	7	12	12	11	
8	8	6	5	6	
9	9	16	18	19	
10	10	14	15	15	
11	11	10	8	9	
12	12	10	12	12	
13	13	14	14	14	
14	14	10	10	11	
15	15	14	16	16	
16	16	14	14	14	
17	17	13	14	14	
18	18	8	10	10	
19	19	6	5	5	

Average Final Grade: 10.2050632911392
Pass Rate (%): 51.8987341772152
Avg Absences: 5.56565656565657
High Absentee Count: 14

Process : Average Final Grade:
=AVERAGEX(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade]))
Pass Rate (%): =DIVIDE(COUNTROWS(FILTER(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade])>10)), COUNTROWS(tbl_Students))
Avg Absences:
=AVERAGE(tbl_Lifestyle[absences])
High Absentee Count:
=COUNTROWS(FILTER(tbl_Lifestyle,tbl_Lifestyle[absences]>20))

Context: Average Final Grade: 10.2051 = consistent with the named range calculation (10.21), confirming both methods agree
Pass Rate: 51.8987% = again consistent, validating the DAX logic
Avg Absences: 5.5657 = rounds to 6 as displayed in the KPI row
High Absentee Count: 14 = 14 students have more than 20 absences, representing a chronic absenteeism concern

The consistency between the DAX measures and the named range formulas from Topic 2 is important, it proves the Data Model relationships are working correctly and the calculations are trustworthy.

CubeFunctions

Power Pivot KPIs	.
Avg Final Grade	10.21
PassRate%	51.90%
Average Absences	6
High Absentees (>20 days)	14

Process :

Average Final Grade:
=CUBEVALUE("ThisWorkbookDataModel","[Measures].[Average Final Grade]")
Pass Rate (%):
=CUBEVALUE("ThisWorkbookDataModel","[Measures].[Pass Rate (%)]")
Average Absences: =CUBEVALUE("ThisWorkbookDataModel","[Measures].[Avg Absences]")
High Absentee Count:
=CUBEVALUE("ThisWorkbookDataModel","[Measures].[High Absentee Count]")

Context: The Power Pivot KPIs table showing identical values (10.34, 51.65%, 6, 14) to both the named range analysis and the dashboard KPI row confirms end-to-end consistency across all three calculation methods used in the workbook.

Conditional and Number Formatting

Standard Guideline Requirement	
Minimal 1 Conditional Formatting	Mandatory
Minimal 1 Number Formatting	Mandatory
Minimal 1 Calculated Fields and Measures	Mandatory

Conditional Formatting

	B	C	D
1	grade_1	grade_2	final_grade
2	5	6	6
3	5	5	5
4	7	8	10
5	15	14	15
6	6	10	10
7	15	15	15
8	12	12	11
9	6	5	6
10	16	18	19

Process & context:

final_grade column:
3-color scale applied (red = low, yellow = mid, green = high) — visible in Image 4 showing red cells for grades 5–6, yellow for 10–11, green for 15–19.
A second rule highlights failing grades (<10) with red font, visible in Image 4 rows 2, 3, 9.
absences column: Data Bars applied to visually indicate absence volume.

Number Formatting

Analysis Named Ranges Directory	Result
Average Final Grade	10.21
Highest Grade	20
Lowest Grade	0
Students Passing (>10)	205
Pass Rate %	51.90%
Average Absences	6
Students with 0 Absences	120
High Absentees (>20 days)	14

Process :

- Grade averages on Analysis sheet formatted to 2 decimal places.
- Pass Rate formatted as Percentage with 2 decimal places (51.65% as seen in Image 3).
- Count values formatted with comma separator.

Calculated Fields & Measure

Average Final Grade: 10.2050632911392
Pass Rate (%): 51.8987341772152
Avg Absences: 5.56565656565657
High Absentee Count: 14

Process :

Average Final Grade:
=AVERAGEX(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade]))
Pass Rate (%): =DIVIDE(COUNTROWS(FILTER(tbl_Students, AVERAGEX(RELATEDTABLE(tbl_Grades), tbl_Grades[final_grade])>10)), COUNTROWS(tbl_Students))
Avg Absences:
=AVERAGE(tbl_Lifestyle[absences])
High Absentee Count:
=COUNTROWS(FILTER(tbl_Lifestyle,tbl_Lifestyle[absences]>20))

Context:

Grade Ratio measure (from Topic 5) referenced in PivotTable 1 alongside Average Final Grade and Pass Rate, satisfying this requirement through the Data Model measures surfaced in the PivotTable. It expresses a student's average final grade as a proportion of the maximum possible score

Mastering Charts



Standard Guideline Requirement	
1 Bar Chart	Mandatory
1 Line or Trend Line Chart	Mandatory
1 Pie or Doughnut	Mandatory
1 Scatter Plot or Combination Chart	Mandatory

Bar Chart

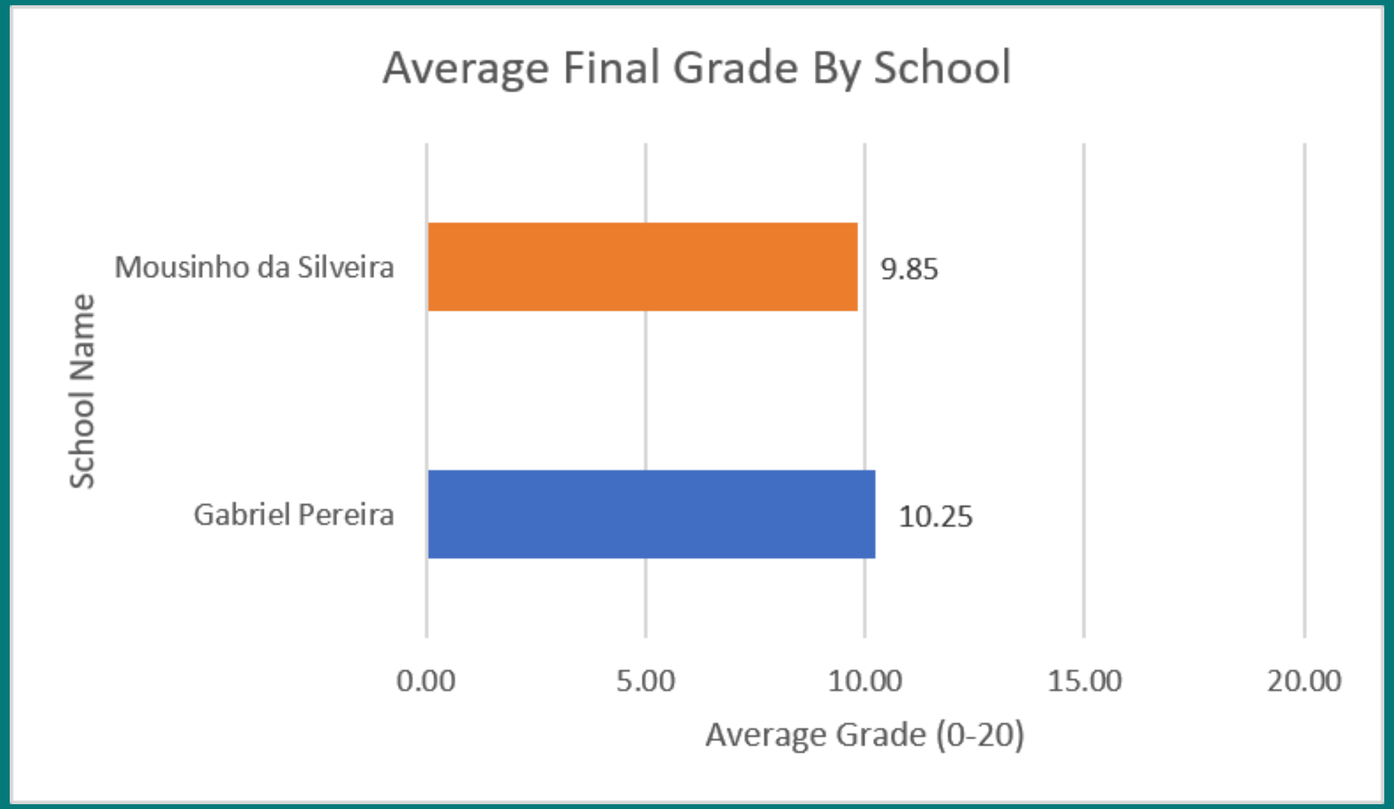
Process :

"Average Final Grade By School":

Gabriel Pereira (10.25) vs Mousinho da Silveira (9.85). Built from AVERAGEIF summary table using school_id as criteria. Shows GP students marginally outperform MS students.

Context:

Gabriel Pereira (10.25) vs Mousinho da Silveira (9.85).
The difference is small, only 0.4 points but GP students are passing on average while MS students are technically just below the threshold. This could reflect differences in teaching quality, student intake demographics, or school resources between the two institutions.



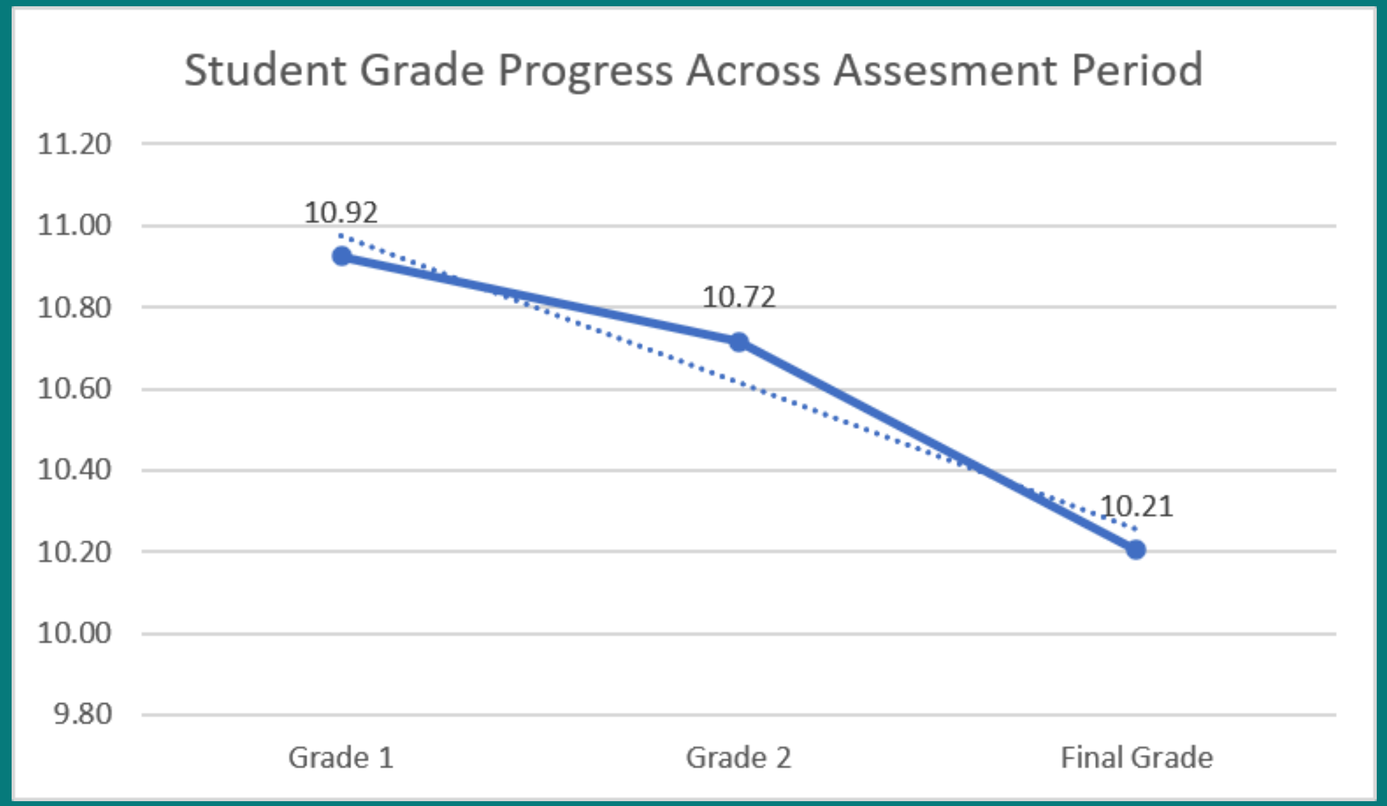
Line Chart

Process :

Shows a consistent downward trend across the academic year. Students perform slightly worse as the year progresses. Linear trendline added (dotted line visible) confirming the downward trajectory.

Context:

The downward trend from Grade 1 (10.92) → Grade 2 (10.72) → Final Grade (10.2) is significant. Students perform progressively worse as the academic year advances. The trendline confirms this is a consistent decline, not random variation. This is a common pattern in schools where final exams are harder or where student fatigue and disengagement increase toward year-end. The practical implication: early intervention based on Grade 1 results could prevent the decline seen by Final Grade.



Pie Chart

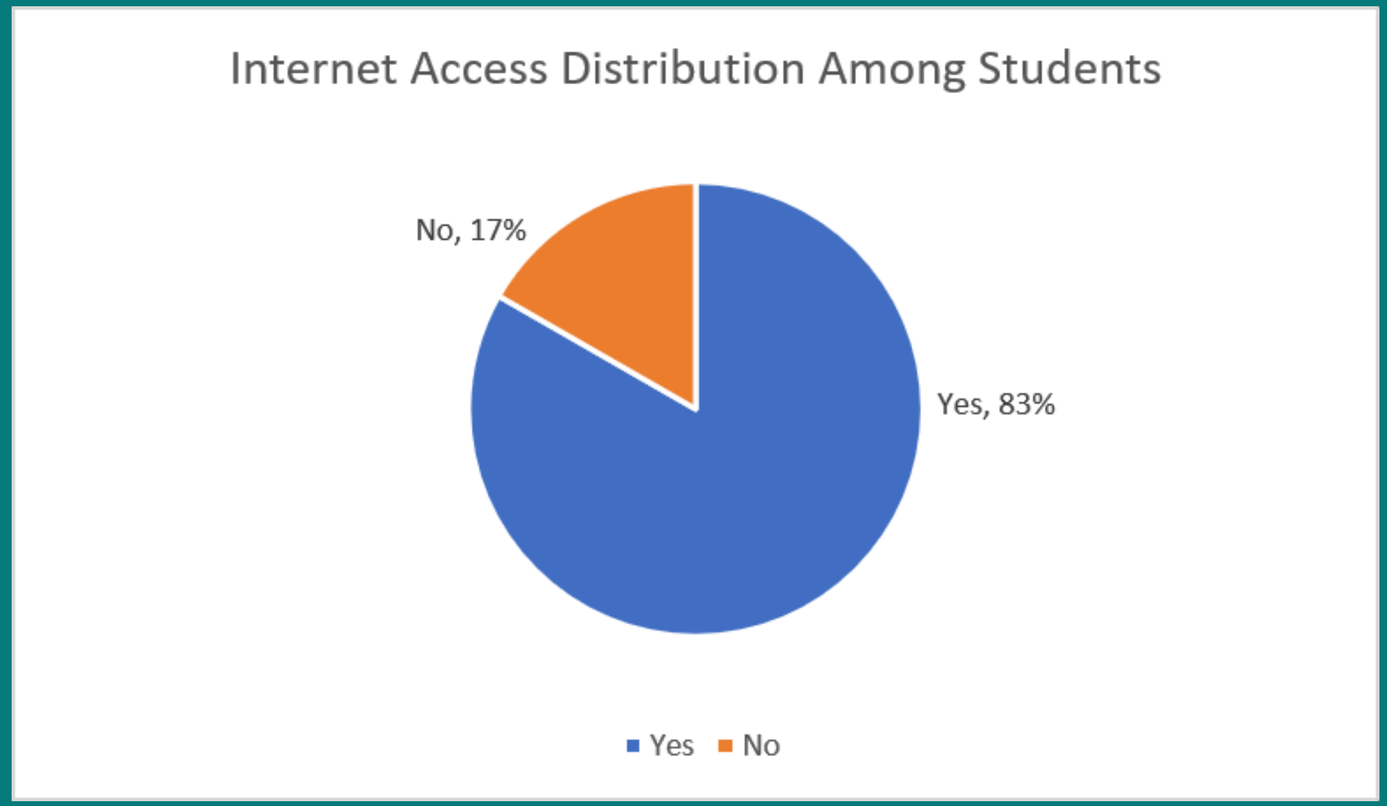
Process :

"Internet Access Distribution Among Students":

83% of students have internet access vs 17% without. Built from COUNTIF on tbl_Lifestyle[internet_access]. Shows strong internet penetration in the student population.

Context:

Internet Access Distribution: 83% yes vs 17% no.
The overwhelming majority of students have internet access, which reflects Portugal's relatively high digital penetration even in secondary schools. The 17% without internet is still a meaningful minority roughly 67 students who may face disadvantages in homework completion and digital learning resources.



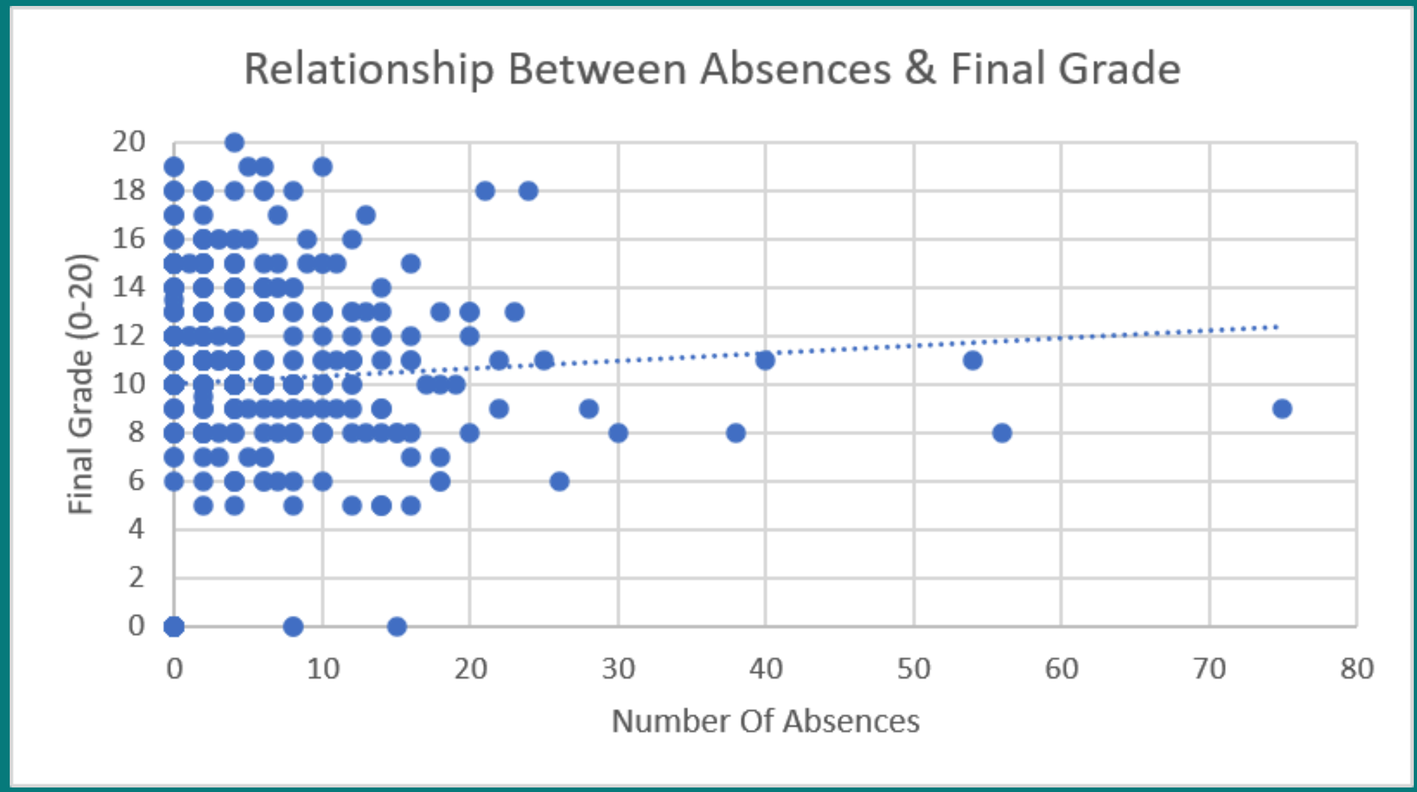
Scatter Chart

Process :

Relationship Between Absences & Final Grade": X-axis: Number of Absences (0–80), Y-axis: Final Grade (0–20). Linear trendline added showing a slightly positive slope.

Context:

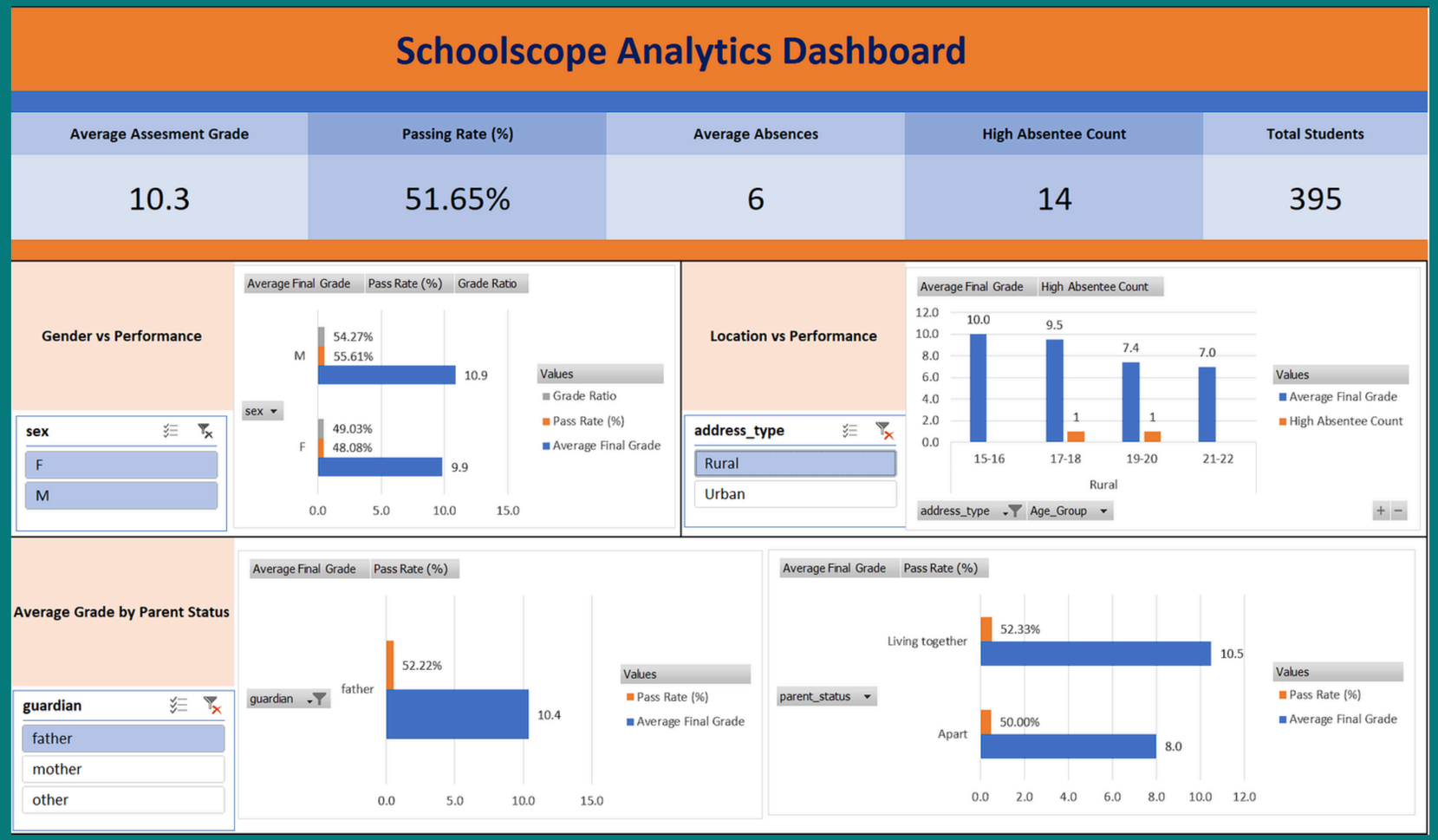
The scatter shows a wide spread with no strong negative correlation the trendline is nearly flat with a very slight positive slope. This is counterintuitive and analytically interesting. It suggests that in this dataset, higher absences do not reliably predict lower grades. Possible explanations include: students who miss school may do so for legitimate medical reasons while maintaining strong academic standing, or the most academically disengaged students simply have low grades regardless of absence count. This finding challenges the assumption that attendance alone determines performance and is worth highlighting explicitly in your report as it demonstrates genuine analytical insight.



Interactive Dashboards

Standard Guideline Requirement	
Minimal 1 halaman dashboard	Mandatory
Menggunakan Pivot Table	Mandatory
Menggunakan Pivot Chart	Mandatory
Menerapkan Sort	Mandatory
Menerapkan Filter	Mandatory
Menerapkan Group	Mandatory
Add Custom Fields	Mandatory
Menggunakan minimal 2 slicer	Mandatory

Dashboard



KPI Row
The five headline metrics tell the school's overall story at both schools.

Gender vs Performance (top-left):
Finding a meaningful gap on the 0–20 scale that could prompt gender-specific academic support programs.

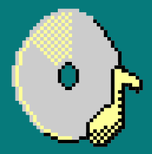
Location vs Performance (top-right):
Finding a meaningful insight how location influences academic outcomes

Guardian Type vs Performance (bottom-left):
Comparison between mother, father, and other guardians will reveal whether parental guardian type influences academic outcomes.

Parent Status vs Performance (bottom-right):
Finding educational research showing family stability as a significant predictor of academic achievement and is the most policy-relevant insight in the entire dashboard.

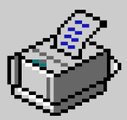








Thank u
<3




11:11PM